



OPEN

A hybrid multi-level ant colony optimization framework for integrated production scheduling and vehicle routing

Mostafa Ashour¹, Ashraf Elsaed¹✉, Yehya Mesalam^{1,2} & Raafat Elshaer^{3,4}✉

This study investigates the integrated scheduling of production and distribution within a time-sensitive supply chain at the operational level. In addition, the study specifically focuses on parallel machine scheduling and vehicle routing problem with time windows (VRP-TW) while considering flexible departure times. A hybrid multi-level optimization (HMLO) framework is developed, decomposing the problem into two primary phases: parallel machine scheduling and distribution scheduling. The initial phase entails the establishment of a comprehensive production schedule, whereas the subsequent phase focuses on segmenting the orders into batches and developing a complete distribution schedule. The framework incorporates the ant colony system (ACS) within the HMLO structure to optimize distribution costs. This is accomplished with both metaheuristics and heuristics to determine the optimal values for the decision variables. Extensive numerical experiments demonstrate that the proposed framework demonstrate that the suggested framework can yield optimal solutions for small-scale instances. Furthermore, it outperforms existing methods, including those utilized in LINGO software, for medium and large-scale instances regarding both convergence and solution quality. For large-scale instances, the proposed method achieves an average improvement of 5 to 30% when compared to LINGO solutions.

Keywords Production and distribution scheduling, Parallel machines, Vehicle routing problem, Hybrid multi-level optimization, Ant colony system

In modern supply chain management, integrating production and distribution phases enhances operational coherence and efficiency. This approach enhances organizational efficiency and ensures an edge in the marketplace. This integration facilitates a seamless process, from manufacturing to delivery to customers.

Industry leaders such as Amazon and Toyota exemplify the significant impact that this integration can have. Through the strategic alignment of their production and distribution processes, they achieve cost savings and establish themselves as leaders in the industry. Integrating production and distribution yields numerous operational benefits. It reduces delivery costs by enhancing operational efficiency. Additionally, it ensures timely deliveries. Amazon's advanced network efficiently links warehouses with delivery routes, facilitating rapid order fulfillment¹. Toyota's renowned Toyota Production System (TPS) maximizes efficiency and minimizes waste through the integration of production with Just-In-Time (JIT) inventory management².

The symbiotic relationship between production and distribution offers various advantages, such as lower costs, increased delivery accuracy, and greater customer satisfaction³. Streamlined operations and synchronized activities help mitigate financial risks linked to delivery delays or premature shipments, thereby promoting stability and trust. The integration of production and distribution is a strategic necessity in the contemporary commercial environment, characterized by agility and a focus on customer needs. Integrating the production and distribution stages can reduce vehicles' time on trips and decrease total lateness⁴. Integrating production and distribution phases allows companies to balance cost and consumer considerations, thereby enhancing customer satisfaction without significantly increasing total operational costs⁵.

¹Industrial Engineering Department, Faculty of Engineering, Zagazig University, Zagazig, Sharkia, Egypt.

²Industrial Engineering Department, College of Engineering, Northern Border University, Arar, Saudi Arabia.

³Industrial Engineering Department, College of Engineering, King Khalid University, Abha, Saudi Arabia.

⁴Center for Engineering and Technology Innovations, King Khalid University, Abha, Saudi Arabia. ✉email: amelsaeed@eng.zu.edu.eg; relshaer@kku.edu.sa

Numerous studies indicate that the integration of production and distribution phases can yield substantial economic advantages, including cost reductions, improved scheduling precision, and enhanced supply chain performance, which encompasses better product quality and customer satisfaction. Bard and Nananukul⁶ note that integrating production and distribution decisions can result in improvements ranging from 10–20% in supply chain optimization. Boudia, et al.⁷ demonstrate that integrating production, inventory, and distribution optimization can result in significant cost reductions, ranging from 3.5 to 27.73%, compared to a traditional two-phase approach. Integrated production–distribution planning can yield up to 20% savings in supply chain costs⁸. He, et al.⁹ examine the potential for an average cost reduction of 16.27% through the integration of 3D printing with just-in-time delivery systems, thereby improving the efficiency of production and distribution scheduling within the supply chain for spare components. Guarnaschelli, et al.¹⁰ indicate that integration in dairy supply chain planning can increase the solution's value by 21.1%.

This study examines the integration of production and outbound distribution stages within the supply chain at the operational level, drawing on practical applications and real-life examples. Specifically, we address a scheduling problem that combines production scheduling for parallel machines with distribution, taking into account VRP-TW and flexible departure times. Initially, customer items must be processed on one of the available identical parallel machines to provide a comprehensive overview of the entire system. Subsequently, the completed items are delivered to designated positions within predetermined delivery time windows utilizing vehicles, considering the flexible departure timings of the vehicles. The primary objective is to reduce the total distribution costs, including penalties for missing delivery time windows and transportation expenses. The paper addresses the following decisions: (1) the assignment of items to machines; (2) the order in which items are processed on machines; (3) the start and end times of each item throughout the production phase; (4) the assignment of items to vehicles; (5) delivery route of each vehicle; (6) departure time for each vehicle; and (7) delivery time of each item.

This study addresses key research questions related to optimizing the integration of production scheduling and vehicle routing with time windows. Specifically, it investigates how a HMLO framework can effectively handle the combined complexities of these two phases. Additionally, the study explores the impact of utilizing the proposed ACS-HMLO framework on solution quality and computational efficiency compared to commercial software (LINGO). Finally, it examines how varying model parameters, such as the number of machines, time window and flexible departure times, affect the framework's performance.

This paper is organized as follows: the literature review is presented in Section “Literature review”, followed by the problem statement and its mathematical model in Section “Problem statement”. Section “Methodology” introduces the HMLO framework. Section “Finally, compute the attractiveness:” presents the numerical experiments and analysis of the framework's effectiveness. Finally, Section “Conclusion and future work” concludes the study and suggests future research directions.

Literature review

Levels of the integration problem

Strategic, tactical, and operational decision levels are involved in distribution and production integration problems. Bilgen and Ozkarahan¹¹ and Fahimnia, et al.¹² offer a comprehensive integration examination encompassing the three levels. At the strategic level, decisions regarding the placement of facilities and the quantity of production are made. Vidal and Goetschalckx¹³ and Sarmiento and Nagi¹⁴ provide a comprehensive discussion and review of these issues. The tactical level encompasses decisions regarding delivery quantities, inventory levels, and the dimensions of manufacturing batches. Díaz-Madroñero, et al.¹⁵ address these decisions. An operational-level schedule is developed to allocate tasks to available resources and establish vehicle routes. Studies by Chen¹⁶ and Wang, et al.¹⁷ examine simple delivery methods, while Moons, et al.³, Berghman, et al.¹⁸, and Elsaed, et al.¹⁹ assess and analyze decision-making related to production and Vehicle Routing Problem (VRP) within the operational distribution framework.

Operational-level decisions are of particular interest of this study due to their immediate influence on resource allocation and vehicle routing, which can lead to significant gains in efficiency and cost reduction. While strategic and tactical decisions play a vital role, the detailed nature of operational decisions provides opportunities for real-time optimization so we will focus on the operational level decisions. Based on review of the literature, particularly studies by Moons, et al.³, Berghman, et al.¹⁸, and Elsaed, et al.¹⁹, the parallel machine environment has been identified as an area requiring further research. This environment, with its complex interaction between production and distribution characteristics, offers a promising avenue for deeper investigation and potential improvements in operational decision-making.

Review of integration between parallel machines and VRP

This section focuses on studies on parallel machines during the production phase and the VRP in the distribution phase. The integration problem can be categorized into two primary types. The initial scenario entails the integration of parallel machines with a single vehicle executing multiple trips. The second scenario entails the integration of parallel machines with an entire fleet of vehicles, where each vehicle completes one trip. The approach of Tavares-Neto and Nagano²⁰ entailed combining a scheduling problem of sequentially setup-dependent parallel machines with a delivery system that consisted of a single vehicle with various routes. A mixed-integer programming (MIP) model and a genetic algorithm (GA) were used to evaluate two algorithms: the iterated greedy (IG) method and the constructive heuristic. Kergosien, et al.²¹ address a scheduling issue that pertains to production and distribution in a chemotherapy environment. Their objective is to reduce the maximum delivery tardiness while considering the chemical stability limits of chemotherapy medications. In order to identify a solution, they compare the results of computational tests conducted using a heuristic based on Benders decomposition to those of a direct exact resolution technique.

The study of Arda, et al.²² details a novel approach to optimize home chemotherapy delivery, integrating production scheduling and multi-trip vehicle routing to minimize the total working time of pharmacists and nurses. A linear program embedded within an adaptive large neighborhood search (ALNS) framework solves the problem, considering stability constraints and time windows. Sugianto and Kim²³ present an integrated scheduling problem for additive manufacturing and delivery, using selective laser melting technology and multi-trip vehicle routing. A mixed-integer linear programming model and an ALNS algorithm are proposed to minimize total completion time. The paper of Yağmur and Kesen²⁴ integrates parallel machine scheduling and vehicle routing, with time windows. Machines operate under discrete speed modes, with lower speeds requiring less energy. Vehicle energy consumption varies based on load. The goal is to minimize the sum of costs related to early/tardy deliveries and production/distribution. A MIP formulation is developed in addition to two metaheuristics, memetic algorithm (MA) and iterated local search (ILS), are proposed for practical-sized instances. Computational results show that ILS performs better in shorter times.

A MILP model is presented for the production scheduling-based routing problem with time window and setup times (PRP-TWST), as developed by Wu, et al.²⁵. A variety of hybrid metaheuristic algorithms, including variable neighborhood search (VNS), particle swarm optimization (PSO), and Cuckoo Search (CS) algorithms, are created and tested in numerical experiments with total delay time set as the objective function. Schubert, et al.²⁶ explore same-day deliveries to meet customer demands, showcasing significant cost savings by integrating the process of order picking with the problem of vehicle routing. A VNS-based approach is introduced and compared to a sequential approach. D. Liu, et al.²⁷ investigate an integrated problem taking into account the due dates of orders. A method is proposed that integrates a GA, the Longest Processing Time heuristic (LPT), and Tabu Search (TS) to address the problem. The proposed framework is based on the decomposition of the problem into three subproblems: vehicle assignment, distribution scheduling, and parallel machine scheduling. Liu, et al.²⁸ investigate the same problem and the same decomposing method, considering the order time window and flexible vehicle departure time. An HMLO framework is suggested that combines GA with the First Fit Decreasing-Longest Processing Time rule (FFD-LPT) and the TS algorithm. Additionally, a method is proposed to determine the optimal departure time for the vehicle efficiently.

Wang, et al.²⁹ solve the integrated production and multiple trips VRP-TW to minimize travel costs and late penalties. They present the “Elastic ρ -Robustness” technique using an enhanced MA. The study of Kesen and Bektaş³⁰ presents a problem of production and distribution, with two variants based on idle waiting in tours, and evaluates the formulations in terms of optimality and computational time, showing slightly different performance metrics between the two cases. Gharaei and Jolai³¹ propose a methodology that includes developing a MILP formulation for the integrated production scheduling and distribution problem, using a branch and price framework with a bees algorithm (BA) for initial column construction. The proposed method is evaluated against established algorithms, including the MILP solver and the branch-and-pricing algorithm. Dayarian and Desaulniers³² address the problem of integrating production routing with time windows in a catering services company. Release times and due dates are considered due to the significance of product freshness. The implementation of an accurate branch-price-and-cut algorithm facilitates cost optimization and adherence to time constraints for small-size instances. Two metaheuristics based on the branch-price-and-cut heuristic have been developed to tackle big-scale instances. Fu, et al.³³ solve a metal packaging production scheduling and VRP to reduce setup and transportation costs. Job splitting and sequence-dependent setup time are permitted. The issue is resolved through the use of a two-phase iterative heuristic and a mathematical model. Zhong and Jiang³⁴ analyze a parallel machine scheduling model that involves batch delivery to two customers. The objective is to minimize the tradeoff between the maximum delivery time of the work and the overall distribution cost. The worst-case performance of a heuristic algorithm with polynomial time complexity is analyzed. Lee, et al.³⁵ outline the difficulties associated with the manufacture and distribution of nuclear medicine, proposing a solution that improves the time and cost aspects. The methodology involves developing a mixed integer program and proposing an ALNS algorithm. The research conducted by Chang, et al.³⁶ focuses on a combined production and distribution scheduling problem. To address this issue, they employ an Ant colony optimization (ACO) heuristic and a dynamic programming technique. The goal is to minimize both the weighted task delivery time and the whole distributing cost. Ullrich³⁷ uses a GA that considers work processing times, delivery time windows, and service times to integrate production and outbound distribution scheduling in order to reduce total tardiness. Using a branch-and-bound strategy and column generation approaches, Chang, et al.³⁸ model the integration issue as a MIP and compare the outcomes. The objective is to minimize the weighted sum of total time and cost associated with task delivery.

Belo-Filho, et al.³⁹ address the production and distribution of perishable products by proposing an ALNS method that utilizes MILP models. The objective is to minimize costs related to setup, production, fixed vehicle expenses, and distance. Amorim, et al.⁴⁰ highlight the significance of lot sizing in the production and distribution planning of perishable goods, taking into account factors such as perishability, customer time windows, and setup times. Farahani, et al.⁴¹ examine the impact of integrating production and distribution planning on enhancing the quality of perishable food through the reduction of delivery time. Iterative solutions are employed to minimize the weighted sum of setup, food decay, and transportation costs. Mathematical models and heuristic solutions are developed through the application of MILP modeling for production scheduling and ALNS for distribution.

Gaps and research opportunity

The summary of relevant literature is shown in (Table 1). We can conclude that there are some gaps in literature: firstly, current research primarily focuses on simpler or more abstract versions of these problems, there is a lack of comprehensive integration of realistic constraints and practical scenarios. Some specific attributes often neglected or not fully integrated include:

	Production attributes						
	Batch production	Processing times	Production cost	Setup times	Setup cost	Production due date	Precedence relationships
Tavares-Neto and Nagano ²⁰	✓			✓			✓
Kergosien, et al. ²¹	✓						
Arda, et al. ²²	✓						
Sugianto and Kim ²³	✓						
Yagmur and Kesen ²⁴	✓						
Wu, et al. ²⁵	✓			✓			
Schubert, et al. ²⁶	✓	✓					✓
D. Liu, et al. ²⁷	✓						
Liu, et al. ²⁸	✓						
Wang, et al. ²⁹	✓						
Kesen and Bektaş ³⁰	✓						
Gharaei and Jolai ³¹	✓						
Dayarian and Desaulniers ³²			✓	✓	✓	✓	✓
Fu, et al. ³³	✓			✓			
Zhong and Jiang ³⁴	✓						
Lee, et al. ³⁵	✓		✓				
Chang, et al. ³⁶	✓					✓	
Ullrich ³⁷							
Chang, et al. ³⁸	✓						
Belo-Filho, et al. ³⁹	✓		✓	✓	✓		
Amorim, et al. ⁴⁰	✓						
Farahani, et al. ⁴¹	✓				✓		
This study	✓						
Distribution attributes							
	Single vehicle	Homogeneous fleet	Heterogeneous fleet	Heterogeneous fleet	Unlimited number	Limited number	Batch delivery
Tavares-Neto and Nagano ²⁰	✓						✓
Kergosien, et al. ²¹	✓						✓
Arda, et al. ²²		✓			✓		✓
Sugianto and Kim ²³	✓	✓				✓	✓
Yagmur and Kesen ²⁴	✓	✓				✓	✓
Wu, et al. ²⁵			✓		✓		
Schubert, et al. ²⁶			✓		✓		
D. Liu, et al. ²⁷			✓		✓		✓
Liu, et al. ²⁸		✓			✓		✓
Wang, et al. ²⁹		✓			✓		✓
Kesen and Bektaş ³⁰		✓			✓		✓
Gharaei and Jolai ³¹		✓			✓		✓
Dayarian and Desaulniers ³²		✓			✓		✓
Fu, et al. ³³			✓				✓
Zhong and Jiang ³⁴		✓			✓		✓
Continued							

	Distribution attributes									
	Single vehicle	Homogeneous fleet	Heterogeneous fleet	Unlimited number	Limited number	Batch delivery	Multiple trips	Delivery due date	Time windows	
Lee, et al. ³⁵		✓		✓					✓	
Chang, et al. ³⁶		✓		✓						
Ullrich ³⁷									✓	
Chang, et al. ³⁸		✓								
Belo-Filho, et al. ³⁹		✓		✓					✓	
Amorim, et al. ⁴⁰		✓		✓					✓	
Farahani, et al. ⁴¹		✓		✓					✓	
This study		✓		✓		✓			✓	
	Distribution Attributes									
	Flexible departure time	Travel times	Variable transportation cost	Fixed transportation cost	Service times	Pickup and delivery	Penalty costs	Tardiness times	Earliness times	
Tavares-Neto and Nagano ²⁰		✓								
Kergosien, et al. ²¹		✓						✓		
Arda, et al. ²²		✓			✓					
Sugianto and Kim ²³		✓								
Yağmur and Kesen ²⁴		✓					✓	✓	✓	
Wu, et al. ²⁵		✓						✓		
Schubert, et al. ²⁶		✓	✓	✓		✓				
D. Liu, et al. ²⁷		✓	✓				✓			
Liu, et al. ²⁸	✓	✓	✓				✓			
Wang, et al. ²⁹		✓	✓		✓		✓			
Kesen and Bektaş ³⁰		✓			✓					
Gharai and Jolai ³¹		✓	✓	✓				✓		
Dayarian and Desaulniers ³²		✓	✓		✓					
Fu, et al. ³³		✓								
Zhong and Jiang ³⁴			✓	✓						
Lee, et al. ³⁵		✓	✓	✓	✓					
Chang, et al. ³⁶		✓	✓	✓						
Ullrich ³⁷		✓			✓			✓		
Chang, et al. ³⁸		✓	✓	✓						
Belo-Filho, et al. ³⁹		✓	✓	✓						

	Distribution Attributes											
	Flexible departure time	Travel times	Variable transportation cost	Fixed transportation cost	Service times	Pickup and delivery	Penalty costs	Tardiness times	Earliness times			
Amorim, et al. ⁴⁰		✓	✓	✓	✓							
Farahani, et al. ⁴¹		✓	✓									
This study	✓	✓	✓				✓	✓	✓			
	Objective											
	Multi objective	Delivery times	Delivery costs	Penalty times	Penalty costs	Makespan	Production costs	Production completion time	Setup cost	Solution method		
Tavares-Neto and Nagano ²⁰						✓				GA,IG and constructive heuristic		
Kergosien, et al. ²¹				✓						Benders decomposition		
Arda, et al. ²²	✓							✓		ALNS		
Sugianto and Kim ²³								✓		ALNS		
Yâğmur and Kesen ²⁴		✓			✓		✓			ILS, MA		
Wu, et al. ²⁵				✓						VNS, PSO, and CS		
Schubert, et al. ²⁶		✓					✓			VNS		
D. Liu, et al. ²⁷		✓			✓					GA		
Liu, et al. ²⁸		✓			✓					GA		
Wang, et al. ²⁹		✓			✓					MA		
Kesen and Bektaş ³⁰				✓						-		
Gharaei and Jolai ³¹	✓	✓		✓						BA		
Dayarian and Desaulniers ³²		✓					✓			Branch-price-and-cut		
Fu, et al. ³³		✓							✓	Iterative heuristic		
Zhong and Jiang ³⁴	✓	✓								Polynomial time heuristics		
Lee, et al. ³⁵		✓					✓			ALNS		
Chang, et al. ³⁶		✓					✓			ACO		
Ullrich ³⁷					✓					GA		
Chang, et al. ³⁸		✓					✓			Column generation		
Belo-Filho, et al. ³⁹		✓					✓			ALNS		
Amorim, et al. ⁴⁰		✓							✓	-		
Farahani, et al. ⁴¹		✓							✓	ALNS		
This study		✓								ACS		

Table 1. The summary of literature. *GA* genetic algorithms, *IG* iterated greedy, *ALNS* adaptive large neighborhood search, *ILS* iterated local search, *MA* memetic algorithm, *VNS* variable neighborhood search, *PSO* particle swarm optimization, *CS* cuckoo search, *BA* bees algorithm, *ACO* ant colony optimization, *ACS* ant colony system.

1. Batch delivery: While some studies consider multiple trips, the specific aspect of batch delivery is not always explicitly addressed.
2. Flexible departure times: Not all models allow for flexible departure times for vehicles, which is a common need in real-world logistics.
3. Penalty costs: The inclusion of penalty costs for tardiness or earliness is not consistently present across the literature.
4. Tardiness and earliness times: Many models do not explicitly incorporate both tardiness and earliness costs/times.

Secondly, methodological Gaps, While various optimization techniques are used, there is room for exploring hybrid approaches, such as combining ACS with other optimization methods and heuristics. Hybrid approaches have the potential to improve the performance and robustness of the solutions. These hybrids can take advantage of the ACS's strengths in exploring solution spaces while using other methods for optimization and addressing complex real-world constraints like parallel machines and VRP-TW. This paper addresses the complex integration of production and distribution scheduling, contributing to the existing literature on the application of ACS to real-world integration issues.

This study is distinguished by its incorporation of real-world attributes like batch delivery, flexible departure time, penalty costs, tardiness and earliness times. This contributes to both practical integration efforts and the theoretical advancement of ACS applications.

The current paper makes contributions that cover some of gaps in the literature. First, we formulate a Mixed-Integer Linear Programming (MILP) model that address the complex issue of integrated production and distribution. The model explicitly incorporates batch delivery, flexible departure times, penalty costs, tardiness, and earliness times. This addresses the gap in current research by providing a more realistic model. Second, we propose a novel hybrid framework that combines ACS with heuristics and procedures to effectively handle the integration problem with realistic constraints.

Problem statement

Efficient integration of production scheduling and vehicle routing is crucial in industries where timely delivery, cost minimization, and resource optimization directly impact operational performance and customer satisfaction. Many real-world supply chain systems face significant challenges in coordinating these interconnected stages, especially under constraints such as limited machine availability, vehicle capacity, and strict delivery time windows^{6,29}. Failure to integrate these phases can lead to increased transportation costs, production delays, and delivery penalties^{31,42}. This study aims to develop a robust, scalable framework that optimally schedules production while synchronizing distribution activities to reduce overall costs associated with transportation and penalty.

Therefore, this study examines a scheduling problem that encompasses production and VRP-TW, which is a coordinated system for production and distribution, with the aim of reducing distribution costs, including transportation expenses and penalties for time window violations. The proposed problem will cover some attributes that considered as gaps in literature.

Problem description

This study examines a scheduling problem that encompasses production and vehicle routing with time windows (IPS-VRTW), as depicted in (Fig. 1). In the production stage, a plant, functioning as either a producer or a supplier, employs comparable machinery to concurrently process items designated for specific customers, with each item linked to a distinct customer. During the distribution phase, completed items are delivered to customer locations using vehicles during prearranged time windows. Each vehicle commences and concludes its journey at the plant depot following deliveries. Penalties are applied for early delivery if it occurs prior to the lower time window limit and for tardiness if the delivery surpasses the upper limit. The task encompasses the following components: (1) assignment of items to machines, (2) determination of the processing order for items on machines, (3) establishment of start and finish timings for each item during production, (4) assignment of items to vehicles, (5) planning of delivery routes for each vehicle, (6) scheduling of departure times for each vehicle, and (7) calculation of delivery times for each item. The aim is to reduce total distribution costs, which include transportation expenses and penalties for time window violations. For simplification, we make the following assumptions: (1) the plant begins in an empty state with scheduling commencing at time 0; (2) all items are predetermined, known, and available for processing at time 0; (3) each item undergoes continuous processing on a single machine without the possibility of splitting, and (4) customer service time is incorporated into travel time.

Mathematical model

The complexity of mathematical models for integrated production and distribution problems, stems from the inherent NP-hard nature of the individual production scheduling and vehicle routing sub-problems, and the added complexity of their integration. Parallel machine scheduling, even without considering distribution, is an NP-hard problem, especially when dealing with parallel machines where processing times vary for each job on each machine⁴³. This is because the problem of minimizing tardiness on a single machine is NP-hard, which makes parallel machine scheduling with tardiness considerations NP-hard as well. The vehicle routing problem (VRP), is also strongly NP-Hard. This is a result of the multiple combinations of routes and delivery quantities, making finding an optimal solution very difficult^{44,45}. Integrating parallel machine scheduling with routing compounds the complexity: Combining these two NP-hard problems makes the integrated production and

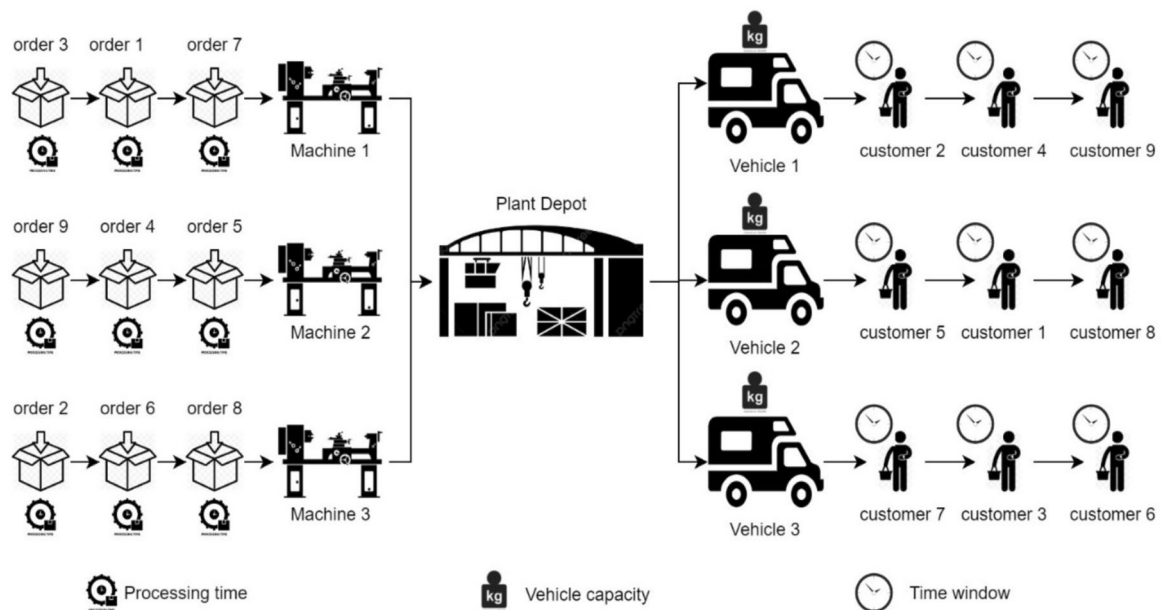


Fig. 1. An example of a coordinated system for production and distribution, including the scheduling of vehicle routes.

distribution problem even more computationally challenging. When these problems are combined, the number of decision variables and constraints increase dramatically, resulting in a more complex mathematical model^{44,45}.

The IPS-VRTW problem involves n items ordered by n customers. In the production stage, k parallel machines are available, and each item needs to be processed by one of the machines without interruption, with the processing time of p_i . The completion time C_i represents the time when item i is completed by one of the available machines. In the distribution stage, h vehicles are available to deliver items, with each vehicle limited to Ca_v , the maximum loading capacity of vehicle v . Each item i has a size of q_i and delivery time of D_i . The cost associated with traveling per time unit is c where d_{ij} is the time taken to travel from customer i to customer j . The costs of violating the time window $[a_i, b_i]$ of item i are e and t , penalty cost per unit time for early and tardy deliveries, respectively.

Below are some more notations that will be defined to introduce the formulation:

Indices

i, j represent customers or items index, where $i, j \in N \cup \{0\}$, where $N = \{1, 2, \dots, n\}$ and 0 denotes the plant depot.

m refers to the available identical parallel machines index, where $m \in M = \{1, 2, \dots, k\}$.

ν denotes the vehicles index, where $\nu \in V = \{1, 2, \dots, h\}$.

Parameters

p_i Time required to process item i .

q_i Size of item i .

d_{ij} Time taken to travel from customer i to customer j .

Ca_v The maximum capacity that vehicle v can hold.

$[a_i, b_i]$ Time window within which item i should be delivered.

LN A sufficiently large value.

c Cost of transportation per unit time.

e Cost incurred for each unit of time as a penalty for early deliveries.

t Cost incurred for each unit of time as a penalty for tardy deliveries.

Decision variables

O_{im} Binary variable takes 1 if item i is allocated to machine m ; otherwise, it takes 0.

X_{ijm} Binary variable takes 1 if item j directly processed after item i on machine m , otherwise it takes 0.

C_i Completion time of item i in the production stage.

l_v Longest completion time of items that will be delivered by vehicle v .

$Y_{i\nu}$ Binary variable takes 1 if item i is delivered with vehicle ν ; otherwise, it takes 0.

$Z_{ij\nu}$ Binary variable takes 1 if item j directly delivered after item i on vehicle ν ; otherwise, it takes 0.

de_v Departure time of vehicle v .

D_i Customer i delivery time.

E_i Time by which item i is delivered earlier than required.

T_i Time by which item i is delivered later than required.

$$\min \left[c * \sum_{i \in N \cup \{0\}} \sum_{\substack{j \in N \cup \{0\} \\ i \neq j}} \sum_{v \in V} d_{ij} Z_{ijv} + e \sum_{i \in N} E_i + t \sum_{i \in N} T_i \right] \quad (1)$$

Subject to:

$$\sum_{m \in M} O_{im} = 1, \forall i \in N \quad (2)$$

$$\sum_{j \in N} X_{0jm} \leq 1, \forall m \in M \quad (3)$$

$$O_{im} = \sum_{j \in N \cup \{0\}} X_{ijm}, \forall i \in N, i \neq j, m \in M \quad (4)$$

$$\sum_{j \in N \cup \{0\}} X_{ijm} = \sum_{j \in N \cup \{0\}} X_{jim}, \forall i \in N, i \neq j, m \in M \quad (5)$$

$$C_j \geq C_i + p_j - LN(1 - X_{ijm}), \forall j \in N, i \in N \cup \{0\}, i \neq j, m \in M \quad (6)$$

$$\sum_{j \in N} Z_{0jv} \leq 1, \forall v \in V \quad (7)$$

$$\sum_{v \in V} Y_{iv} = 1, \forall i \in N \quad (8)$$

$$Y_{0v} \geq Y_{iv}, \forall i \in N, v \in V \quad (9)$$

$$Y_{iv} = \sum_{j \in N \cup \{0\}} Z_{ijv}, \forall i \in N, i \neq j, v \in V \quad (10)$$

$$\sum_{j \in N \cup \{0\}} Z_{ijv} = \sum_{j \in N \cup \{0\}} Z_{jiv}, \forall i \in N, i \neq j, v \in V \quad (11)$$

$$\sum_{i \in N} q_i Y_{iv} \leq Ca_v, \forall v \in V \quad (12)$$

$$l_v \geq C_i - LN(1 - Y_{iv}), \forall v \in V, i \in N \quad (13)$$

$$de_v \geq l_v, \forall v \in V \quad (14)$$

$$D_i = de_v + d_{0i} - LN(1 - Z_{0iv}), \forall i \in N, \forall v \in V \quad (15)$$

$$D_j = D_i + d_{ij} - LN(1 - Z_{ijv}), \forall v \in V, \forall i, j \in N, i \neq j \quad (16)$$

$$E_i \geq a_i - D_i, \forall i \in N \quad (17)$$

$$T_i \geq D_i - b_i, \forall i \in N \quad (18)$$

$$C_i, D_i, E_i, T_i \geq 0, \forall i \in N \quad (19)$$

$$de_v, l_v \geq 0, \forall v \in V \quad (20)$$

$$Y_{iv}, O_{im}, X_{ijm}, Z_{ijv} \in \{0, 1\}, \forall v \in V, m \in M, \forall i, j \in N \cup \{0\}, i \neq j \quad (21)$$

The objective function (1) aims to minimize the total distribution cost, encompassing transport costs and penalties for violations of delivery windows. Constraints (2) to (6) delineate the production stage; Constraint (2) ensures that each item is assigned to one machine exclusively. Constraint (3) ensures that each machine is limited to producing a single item at any given time. The elements are arranged sequentially, with each item positioned either before or after another, as specified by constraints (4) and (5). The determination of an item's completion time relies on its processing time and the completion time of the preceding item, as outlined in constraint (6). Constraints (7) to (18) delineate the distribution stage; constraint (7) stipulates that only one item may be designated as the first order after the depot for each utilized vehicle. Constraints (8) and (9) guarantee that each item is assigned to a single vehicle and that every vehicle is required to visit the plant depot. Constraints (10) and (11) specify that each customer may be served by a single vehicle only. No vehicle's load can exceed its capacity, as stipulated in constraint (12). Constraint (13) stipulates that the makespan for orders delivered by each vehicle must encompass the completion time of those orders. Constraint (14) stipulates that the departure time for each

vehicle must not be earlier than the makespan of the orders it delivers. Constraints (15) and (16) calculate the delivery times based on routing and departure decisions by adding the vehicle's departure time to the travel time from the depot to the order location for the first order. For subsequent orders, the delivery time is calculated by adding the travel time between the two orders to the delivery time of the previously delivered. Constraints (17) and (18) calculate the early and tardy times for each order. Constraints (13)–(18) collectively establish a temporally consistent delivery chain, where: (13) ensures the vehicle's latest item completion time bounds its assigned items' production times; (14) anchors vehicle departure de_v to l_v , preventing premature departures; (15)–(16) deterministically calculate delivery times using travel durations (d_{ij}) while the big- M term (LN) deactivates constraints for inactive routes to avoid artificial infeasibility; and critically, (17)–(18) implement *soft time windows* where E_i and T_i quantify earliness/tardiness deviations as penalty terms in the objective function (1), ensuring all solutions remain feasible. Violations of time windows incur penalties in the objective function but do not invalidate feasibility. This cascaded design ($C_i \rightarrow l_v \rightarrow de_v \rightarrow D_i \rightarrow E_i/T_i$) guarantees operational coherence while allowing penalized schedule flexibility. Finally, constraints (19) to (21) are constraints on non-negativity and binary variables.

Methodology
HMLO framework based on ACS

The ACS is a metaheuristic algorithm inspired by the foraging behavior of ants, extending the foundational ant system (AS) algorithm^{46–48}. AS, the first ACO algorithm, was developed in 1991, and it introduced the idea of using artificial ants to build solutions for combinatorial problems through a pheromone-based communication system^{46,47}.

ACS was subsequently developed to enhance the performance of AS, particularly for the Traveling Salesman Problem (TSP), by including a state transition rule to balance exploration and exploitation of the search space, a global updating rule that focuses on the best ant tour, and a local pheromone updating rule⁴⁶. The main benefits of applying ACS include its ability to enhance exploration around good solutions with local pheromone updates and focus on the best solutions with global updates, a cooperative search using distributed memory, and adaptability by transferring knowledge from previous optimization runs^{49,50}. ACS can be applied to different combinatorial problems by using a graph representation and a heuristic that guides the construction of solutions^{46,51}. While several ACO variants exist, such as Max–min ant system (MMAS), rank-based ACO, and Elitist Ant System, the ACS was selected for this study due to its balanced control over exploration and exploitation, which is particularly valuable in integrated optimization problems involving multiple decision layers and constraints. Compared to classical ACO, ACS incorporates a pseudo-random proportional state transition rule, global updating based on the best solution, and local pheromone updates, enabling it to avoid premature convergence while intensifying search near promising regions. Compared to MMAS, which emphasizes global search with stricter pheromone bounds, ACS allows for more adaptive learning in dynamic problem landscapes. Furthermore, ACS has demonstrated robust performance in both scheduling and routing domains independently, which supports its applicability to their integration. These design features make ACS particularly well-suited for problems with complex interdependencies, such as synchronizing production and delivery decisions under time window constraints. A comparative summary of ACO variants is presented in (Table 2).

In the realm of production scheduling, ACS applied to the job-shop scheduling problem⁴⁷. ACS has been utilized to solve parallel machine scheduling problems^{43,55}, as well as job-shop scheduling⁵⁶. In vehicle routing, ACS has been applied to both the VRP and the VRPTW^{47,49,50,57}. The ACS algorithm is also used for the integration of production and distribution scheduling, using separate pheromone matrices for production and distribution, also considering batch delivery and multi-factory supply chain problems with routing^{56,58}. These

Algorithm					
Feature	Ant system (AS)	Elitist ant system (EAS)	Rank-based ACO	Max–min ant system (MMAS)	Ant colony system (ACS)
First proposed	1991	1991	1997	1998	1997
Reference	Dorigo, et al. ⁵²	Dorigo, et al. ⁵²	Bullnheimer, et al. ⁵³	Stützle and Hoos ⁵⁴	Dorigo and Gambardella ⁴⁶
Pheromone update rule	All ants	All ants + elitist ants	Ranked ants only	Only best or iteration-best ant	Only global-best ant + local update
Pheromone update scope	Global	Global	Global	Global	Global + Local
Exploration–exploitation	Balanced but less controlled	More exploitation (elitism bias)	High exploitation (top-ranked ants)	Strong exploitation, bounded pheromone	Well-balanced via pseudo-random rule
Use of local update	X	X	X	X	✓ Local pheromone update after each move
Candidate list (lookahead)	X	X	X	Optional	✓ Encourages local decision efficiency
Pheromone bounds	X	X	X	✓ τ_{min} , τ_{max} constraints	X (but implicitly stable via updates)
Search diversification	Moderate	Low	Low	Strong	✓ Strong via state transition rule
Performance in complex problems	Moderate	Moderate	Good (with careful tuning)	Good (with tuning)	✓ Robust and adaptive
Parameter sensitivity	High	High	High	Very High	✓ Moderate and easier to calibrate
Applicability to integrated problems	Limited	Limited	Rare	Moderate	✓ Frequently used in integrated scheduling & routing

Table 2. Comparison of ACO variants and their suitability for integrated scheduling.

applications of ACS demonstrate its effectiveness in solving complex optimization problems in production, vehicle routing and integration of the two^{44,58}. Based on the comparison above in (Table 2), ACS offers several critical advantages that make it well-suited to solving integrated production and distribution problems:

- It combines global and local pheromone updates, enhancing both solution quality and convergence speed.
- Its pseudo-random state transition rule promotes effective balance between intensification and diversification.
- The candidate list mechanism reduces computational complexity during solution construction.
- ACS has a track record of success in related applications (e.g., VRP, parallel machine scheduling, integrated logistics).
- Compared to MMAS and Rank-Based ACO, ACS is simpler to implement and less sensitive to parameter tuning, making it a practical and reliable choice.

The HMLO framework is a valuable approach for addressing complex integrated production and distribution problems due to its ability to decompose the problem into smaller, more manageable sub-problems⁴². By breaking down the larger problem, HMLO facilitates more effective solutions compared to addressing the entire problem simultaneously⁴². Previous studies have demonstrated that hybrid metaheuristic approaches significantly improve solution efficiency in integrated scheduling and routing problems. The HMLO framework proves to be a robust and efficient method for tackling complex integrated production and distribution scheduling problems^{42,59–61}.

In our IPS-VRTW problem, Ten decision variables were examined. The determination of distribution scheduling within the operational framework is intricately connected to the decision about production scheduling. In other words, the manner and timing of order production determine the manner and timing of their distribution. Upon determining the variables (O_{im} , X_{ijm} , Y_{iv} , de_v , and Z_{ijv}), we can efficiently compute the values of the five variables (C_i , l_v , D_i , E_i , and T_i). Our emphasis should be on optimizing the initial five variables. Consequently, we decompose the problem into three subproblems. As shown in Fig. 2, we first tackle a complete production scheduling sub-problem in Production-level optimization where the orders are assigned to machines, and their processing sequences are determined considering processing time, traveling times, and time windows of orders (i.e., determining O_{im} and X_{ijm}). Notably, this is where distribution attributes impact production scheduling since the time windows and traveling times of the orders are considered. Then, in the distribution-level optimization process, we determine Y_{iv} by dividing orders into delivery batches according to the completion time in the production stage. Thus, the production stage affects the distribution stage by providing the completion times which dictates the formation of delivery batches. Then, each batch relates to a vehicle for which a departure time (de_v) can be determined. Finally, the distribution scheduling sub-problem is tackled by determining routes (Z_{ijv}) and optimal departure times for vehicles.

Based on Fig. 2, we construct the ACS-HMLO illustrated in Fig. 3, integrating production-level optimization with a distribution-level optimization process. ACS, a probabilistic search method, utilizes the pheromone model to sample the search space.

This research involves multiple ants generating solutions probabilistically, guided by a designed pheromone model in each iteration. The optimal approach involves updating the pheromones prior to the commencement of the subsequent iteration. Two matrices are established: one for production and another for distribution, specifically for pheromone updates. The ACS-HMLO algorithm framework, depicted in Fig. 3, involves the initialization of algorithm parameters, followed by the development of a production schedule utilizing transition rules that account for processing times, travel times, and order time windows (steps 1 and 2). Following the local update of the production pheromone matrix in Step 3, delivery batches are generated and allocated to vehicles according to the production schedule in Step 4. Steps 5 to 10, represent the ACS algorithm used to find the best routes for the proposed distribution batches in the previous step. A distribution schedule is constructed by the distribution transition rule based on a new ant colony. Subsequently, the distribution pheromone matrix is locally updated following the construction of a distribution schedule. This process continues until all distribution's ants find feasible solutions for distribution. Afterwards, the distribution pheromone matrices are globally updated using the optimal solution obtained during the iteration. The ACS algorithm proceeds until

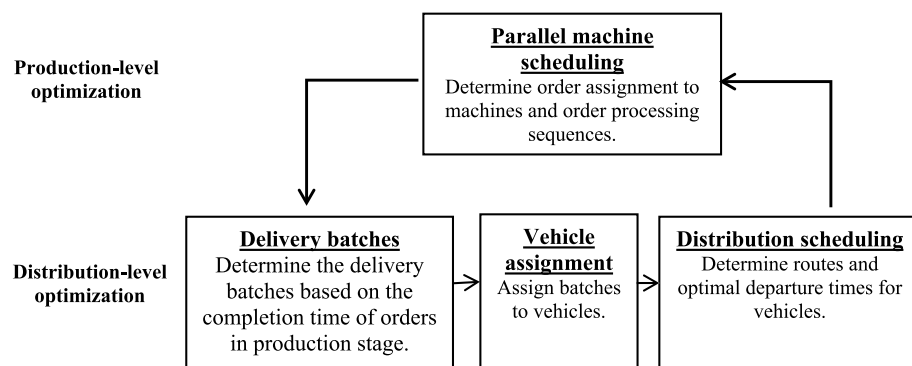


Fig. 2. Decomposing the IPS-VRTW problem into subproblems.

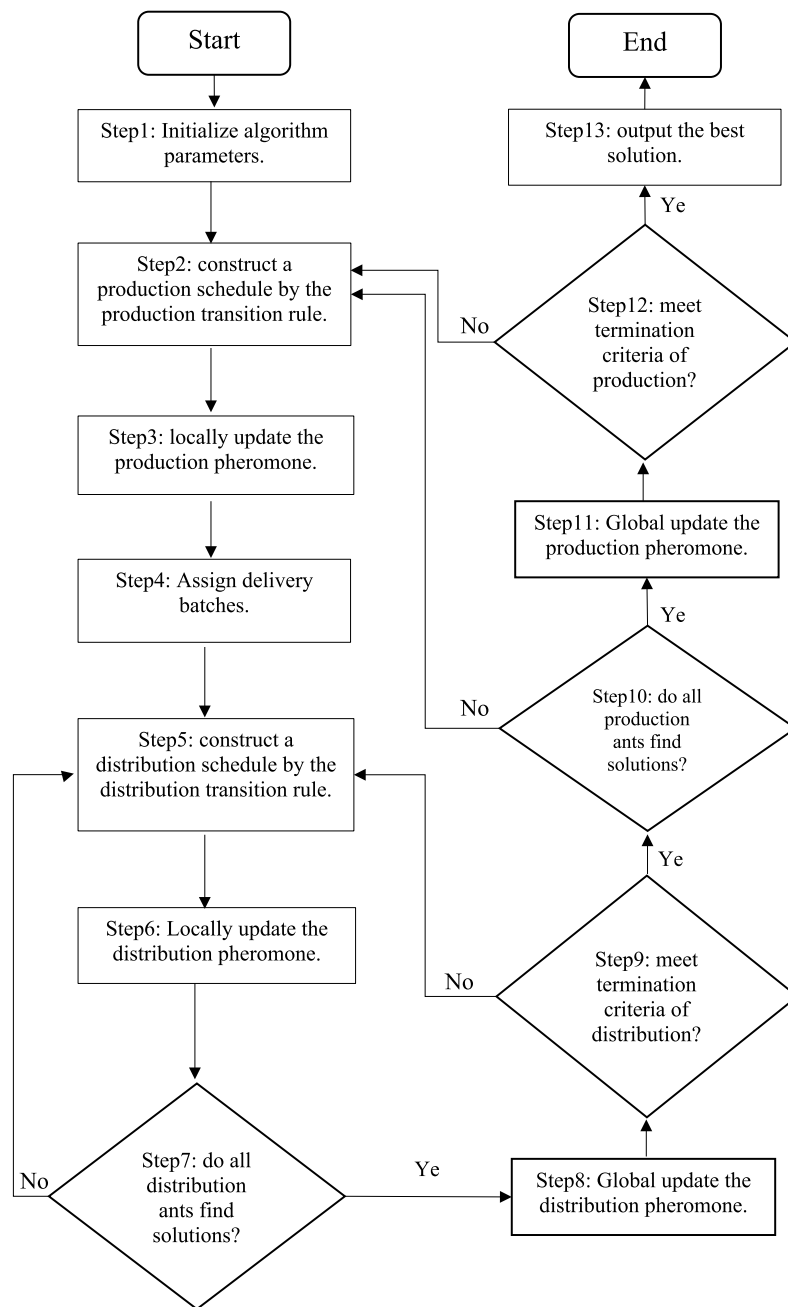


Fig. 3. The ACS-HMLO flowchart.

the maximum iteration limit in the distribution is achieved, optimizing the distribution schedule to finalize the remaining steps. From step 10, the production ant colony continue and finds a new solution until all production ants find one. Then the pheromone matrix of production is globally updated based on the best ant found. Once the production reaches the maximum number of iterations, the ACS-HMLO algorithm terminates, and the optimal solution identified is considered.

Construction of schedules

The construction of ant paths consists of three sequential parts. Initially, the production schedule is determined. Subsequently, the delivery batches are arranged. Finally, the route for each vehicle is determined.

Parallel machines scheduling

The scheduling of identical parallel machines represents a significant challenge in manufacturing and operations research. Job scheduling involves the allocation of tasks to a group of machines with identical capabilities to optimize specific metrics, such as minimizing total task completion time, reducing overall completion duration, or achieving an equitable distribution of workload among the machines. The ACS is a highly efficient metaheuristic

that draws inspiration from the foraging behavior of ants. ACS is widely recognized for its robustness and adaptability in solving a broad range of optimization problems. The behavior of ants foraging for food serves as inspiration for ACS. Ants release pheromones on the paths they travel, and the level of pheromones affects the likelihood of other ants choosing the same path. This positive feedback process enables the colony to identify optimal or almost optimal solutions to intricate challenges. Ants utilize their past experiences to identify optimal trails and simultaneously explore uncharted paths in search of improved solutions. These strategies are referred to as 'exploitation' and 'exploration', respectively. Exploitation aims to enhance previously identified solutions; however, it may result in convergence to a local optimum. Exploration allows ants to move beyond local optima and pursue improved solutions in unexplored regions. The state transition rule developed by Dorigo and Gambardella⁴⁶ and Gambardella, et al.⁵⁷ is used to decide the following order to process.

In the proposed ACS-based framework, the attractiveness η_{ij} represents the desirability of transitioning from order i to order j during route construction. This measure considers both temporal and historical characteristics of order j , ensuring that the ant's decision process is informed by time feasibility, delivery priority, and solution learning. The attractiveness η_{ij} is determined by considering the processing time p_j , the travel time d_{ij} between order i and j , the time window $[a_j, b_j]$ associated with order j , and the frequency IN_j of order j not being included in previous problem solutions.

The calculation steps, explained by numerical example in Supplementary Note 1 online, for determining η_{ij} are:

1. Calculate the delivery time for order j :

$$\text{delivery.time}_j \rightarrow \max\{\text{current.time}_k + d_{ij}, b_j\} \quad (22)$$

2. Compute the time difference between orders i and j :

$$\text{time}_{ij} \rightarrow \text{delivery.time}_j - \text{current.time}_k \quad (23)$$

3. Determine the initial time estimation measure between orders i and j :

$$\text{time.estimation}_{ij} \rightarrow \Delta \text{time}_{ij} \times (e_j - \text{current.time}_k) \times \text{processing.time}_j \quad (24)$$

4. Adjust the time estimation measure based on the frequency IN_j of order j not being included in previous solutions:

$$\text{time.estimation}_{ij} \rightarrow \max\{1.0, \text{time.estimation}_{ij} - IN_j\} \quad (25)$$

5. Finally, compute the attractiveness η_{ij} :

$$\eta_{ij} \rightarrow \frac{1}{\text{time.estimation}_{ij}} \quad (26)$$

In ACS, m ants construct full machines schedules simultaneously, where m is a predefined parameter. Each ant is allocated a starting order at random and is required to develop an entire schedule. Each schedule is incrementally built as ants iteratively add orders until all have been processed. When ant k is at order i , it probabilistically selects the next order j from the set of feasible orders N_i^k (orders that still need to be processed).

The probabilistic selection rule is as follows: with probability q_0 , the ant chooses the order j that has the highest $\tau_{ij} \cdot [\eta_{ij}]^\beta$, $j \in N_i^k$ (exploitation). Otherwise, with probability $1 - q_0$, the next order j is chosen with a probability P_{ij} proportional to $\tau_{ij} \cdot [\eta_{ij}]^\beta$, $j \in N_i^k$ (exploration, Eq. 27).

$$P_{ij} = \begin{cases} \frac{\tau_{ij} \cdot [\eta_{ij}]^{\beta_s}}{\sum_{l \in N_i^k} \tau_{il} \cdot [\eta_{il}]^{\beta_s}}, & j \in N_i^k \\ 0, & \text{otherwise} \end{cases} \quad (27)$$

where β and q_0 are parameters: β determines the relative importance of the heuristic value, while q_0 influences the balance between exploitation and exploration. $s \in \{p, d\}$ as β_s will be β_P in the production stage and β_d in the distribution stage.

Local and global updates are performed to pheromone trails in ACS. Local updating occurs during the solution construction phase, while global updating takes place at the end of this phase. Locally, the desirability of links is dynamically adjusted: each time an ant traverses a link, the pheromone level on that edge decreases, reducing its attractiveness. Conversely, global updating intensifies the search around the best-found solution. The pheromone global update rule is given by:

$$\tau_{ij} = (1 - \rho) \cdot \tau_{ij} + \rho / J_\Psi^{gb} \quad (28)$$

where ρ ($0 \leq \rho \leq 1$) is a parameter, and J_Ψ^{gb} is the best total cost of current iteration. Upon completion of the construction phase, each iteration ends with this global update.

Local updating, applied when an ant moves from order i to order j , decreases the pheromone level on link (i, j) according to:

$$\tau_{ij} = (1 - \rho) \cdot \tau_{ij} + \rho \cdot \tau_0 \quad (29)$$

where τ_0 is the initial pheromone level, determined by: $\tau_0 = \frac{1}{n}$ as n is the number of orders to be processed. This local updating strategy reduces the attractiveness of frequently used links, promoting the exploration of new links and enhancing solution diversity.

After each ant sorts all orders, the steps of LPT algorithm are used with some modifications to distribute orders between available machines as follow:

Step 1: Use the sorting of each ant.

Step 2: Assign orders to Machines: Iteratively assign each order to the machine that currently has the least total workload (i.e., the machine that will finish its as-signed orders the earliest).

Step 3: Continue Until All Orders Are Assigned: Repeat the process until all orders are assigned to machines.

Step 4: Output the Schedule: The result is a schedule where jobs are distributed across machines according to According to ants' preferences.

Allocation of distribution batches

As the algorithm progresses, the subsequent step involves selecting the orders to be included in a batch and estimating the departure time of the vehicle transporting the batch from the depot. This procedure is simplified using a straightforward heuristic that divides orders into batches based on the time required for the completion of each order. Orders are initially organized according to their completion times, following the First-In-First-Out (FIFO) sequence. This sequence continues until all the orders have been completed. In FIFO sequence, let us designate the order that terminates in the position of i^{th} position ($i \in \{1, 2, \dots, n\}$). The jobs are assigned to the vehicles based on established criteria. This heuristic can be illustrated through a numerical example involving five orders processed on two machines, which will be delivered by two vehicles. The completion times are as follows:

- Before sorting: completion.times = [2, 5, 1, 5, 7], orders = [1, 2, 3, 4, 5]
- After sorting: completion.times = [1, 2, 5, 5, 7], orders = [3, 1, 2, 4, 5]
- Following the sorting of the orders, we categorize them into batches in several attempts as follows:

1 $\rightarrow$ [3], [1, 2, 4, 5]
 2 $\rightarrow$ [3, 1], [2, 4, 5]
 3 $\rightarrow$ [3, 1, 2], [4, 5]
 4 $\rightarrow$ [3, 1, 2, 4], [5]
 5 $\rightarrow$ [3, 1, 2, 4, 5], [].

In the previous example results, five collections of batches require evaluation to identify the optimal collection for consideration. Subsequently, we determine the optimal route and departure time for each batch, assessing each collection of batches accordingly.

Distribution scheduling

Once all orders are assigned to vehicles, the optimal route for each vehicle is determined based on the distribution ACS, and the optimal departure time is calculated. The ACS in the distribution stage follows the same steps as the ACS in the production stage, except that processing time is excluded from Eq. (24). In order to determine the optimal time for departure, we use the procedure developed by Liu, et al.²⁸, which involves the following steps:

Step 1: Use the delivery route to determine the actual delivery time D_i for each item.

Step 2: Sort the items into three sets: L for late deliveries, E for early deliveries, and S for on-time deliveries using the delivery times from Step 1.

Step 3: Compute $\rho = \left| E \right| - \frac{t}{e} \left| L \right|$ as t and e are the costs of tardiness and earliness.

Step 4: If $\rho \leq 0$, the optimum departure time d_v^* will be the same as makespan f_v of all items delivered by vehicle v .

Step 5: If $\rho > 0$, determine d_v^* as follows:

Step 5.1: Initialize $d_v = f_v$.

Step 5.2: Find the set of viable solutions H as $H = \{\Delta d_v \mid \Delta d_v \in A \cup B, \Delta x_1 \leq \Delta d_v \leq \Delta e_2\}$

$A = \{a \mid a = l_i - D_i, i \in E\}$ as E the set of early orders

$B = \{b \mid b = u_j - D_j, j \in G\}$ as G the set of timely delivered orders

$\Delta e_1 = \min A, \Delta e_2 = \max A, \Delta s_1 = \min B, \Delta x_1 = \min(\Delta e_1, \Delta s_1)$

Step 5.3: Locate the lowest possible value $\Delta d_v^{\min} = \min H$ in set H .

Step 5.4: Update $d_v = d_v + \Delta d_v^{\min}$, recalculate H , and compute ρ again.

Step 5.5: If $\rho < 0$, set $d_v^* = d_v - \Delta d_v^{\min}$; if $\rho = 0$, set $d_v^* = d_v$; otherwise, go to Step 5.3.

Figure 4 shows two scenarios that exemplify the implementation of the optimal departure time procedure. The scenarios pertain to the delivery routes of a vehicle originating from a depot (denoted as '0') to multiple customers with specified delivery times and time windows. Additionally, Table 3 depicts the results of applying the optimal departure procedure to the two scenarios. In the first scenario, the initial delivery times were 18, 24, 27, 41, and 47, leading to one late item (8), two early items (1 and 5), and two timely items (4 and 7). With tardiness and earliness costs both equal to 1, the initial ρ value was 1. Consequently, Step 5 was initiated. The sets A and B were determined to be {1, 3} and {1, 2}, respectively, yielding $\Delta e_1 = 1, \Delta e_2 = 3, \Delta s_1 = 1$, and $\Delta x_1 = 1$. The set H was {1}, and the minimum value $\Delta d_v^{\min} = 1$. Updating the departure time d_v by adding

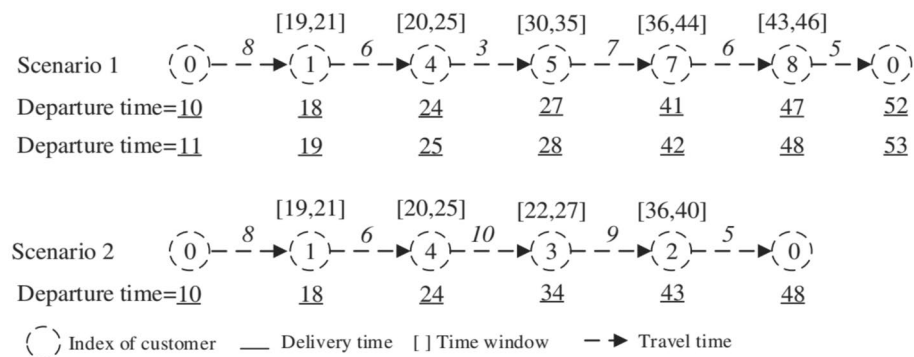


Fig. 4. Two scenarios of vehicle routes²⁸.

	Initial setup	Procedure application	Updated delivery times	Conclusion
Scenario 1	- o Delivery times: 18, 24, 27, 41, 47 - o Late items $L = \{8\}$ - o Early items $E = \{1, 5\}$ - o Timely items $G = \{4, 7\}$ $t = e = 1$ $\rho = \left E \right - \frac{t}{e} \left L \right = 1$	- o Step 5 is initiated - Because $\rho > 0$ $A = \{1, 3\}$ $B = \{1, 2\}$ $\Delta e_1 = \min A = 1$ $\Delta e_2 = \max A = 3$ $\Delta s_1 = \min B = 1$ $\Delta x_1 = \min (\Delta e_1, \Delta s_1) = 1$ - o $H = \{1\}$ $\Delta d_v^{\mu in} = \min H = 1$ - o Updated $d_v = d_v + \Delta d_v^{\mu in} = 11$	- o New delivery times: 19, 25, 28, 42, 48 - o Late items $L = \{8\}$ - o Early items $E = \{5\}$ - o Timely items $G = \{1, 4, 7\}$ $\rho = 0$	o Procedure terminates with $d_v^* = 11$
Scenario 2	- o Delivery times: 18, 24, 34, 43 - o Late items $L = \{3, 8\}$ - o Early items $E = \{1\}$ - o Timely items $G = \{4, 7\}$ $t = e = 1$ $\rho = \left E \right - \frac{t}{e} \left L \right = -1$			o Since $\rho < 0, d_v^* = 10$

Table 3. Results of applying optimal departure procedure on two scenarios.

this minimum value resulted in new delivery times of 19, 25, 28, 42, and 48, with one late item (8), one early item (5), and three timely items (1, 4, 7). With ρ recalculated to 0, the procedure terminated with an optimal departure time d_v^* of 11. In the second scenario, the initial delivery times were 18, 24, 34, and 43, resulting in two late items (3, 8), one early item (1), and two timely items (4, 7). With both tardiness and earliness costs established at 1, the initial ρ value was -1. Given that ρ was less than or equal to 0, the procedure concluded promptly, establishing the optimal departure time d_v^* as 10. These results demonstrate the procedure's effectiveness in optimizing departure times to reduce costs related to late and early deliveries, thereby maximizing the timely delivery of items.

Computational experiments

The numerical results for a collection of problem cases that were evaluated on an Intel i5-12500H CPU with 8 GB of RAM are summarized below.

Design of test instances

Instances are generated utilizing the methods proposed by Ullrich³⁷ and Liu, et al.²⁸. The processing time p_i is from $U[1, \theta]$. The travel time r_{ij} between destinations i and j is from $U[1, \theta h/k]$. The distance traveled, r_{ij} , between orders i and j is from $U[1, \theta h/k]$. The lower time window of customer i is calculated as the sum of its processing time p_i , the travel time r_{0i} from the plant to the customer, and a random integer π_i generating from $U[0, \lfloor \alpha_1 \theta n (k + h) \rfloor]$, giving $a_i = p_i + r_{0i} + \pi_i$. The upper bound b_i is the lower time window plus a random integer number ε_i from $U[0, \lfloor \alpha_2 \theta \rfloor]$, resulting in $b_i = a_i + \varepsilon_i$. For generating test instances, parameters $\theta, \alpha_1, \alpha_2$ are set to 100, 0.5, and 0.5, respectively.

Algorithm parameter setting

The most suitable values for each parameter were determined through hill climbing algorithm. The values identified as the optimal balance between computational time and solution quality were utilized in all subsequent experiments of this study. These values are number of runs=10, $\beta_P = 1$, $\beta_d = 0.8$, $\rho = 0.1$ for production and distribution, $q_0 = 0.7$ for production and distribution, $\tau_0 = 0.001$ for production and distribution. The distribution costs and penalty costs for each unit of time are set to $c=2$, $e=1$ and $t=3$. The number of ants and iterations is determined by the problem size, which can be founded in the given link of results in the following

Section. The hill climbing algorithm along with parameters initial values, corresponding levels and best-found values are reported in details in Supplementary Note 2 online.

Performance measurement

To evaluate the effectiveness of the ACS-HMLO framework in solving the distribution cost optimization problem, the relative difference (*RD*) is used as a performance metric. The optimal distribution cost (*F*^{*}) is determined using the LINGO optimization solver, which provides the best-known solution for each instance. The least distribution cost (*F*) is obtained from the ACS-HMLO framework, representing the solution derived from the heuristic approach. The relative difference (*RD*) is calculated as follows: $RD = \frac{F - F^*}{F^*} * 100\%$. The *RD* metric quantifies the deviation of the ACS-HMLO solution from the optimal solution. A lower *RD* value indicates that the heuristic approach produces a near-optimal solution, while a higher *RD* value suggests a greater gap between the heuristic and optimal results. This measurement provides insight into the accuracy and efficiency of the ACS-HMLO framework in minimizing distribution costs compared to the exact optimization method.

Experimental results and discussions

To provide a clearer understanding of the scalability of the proposed ACS-HMLO framework, the test instances were categorized into three levels based on the number of customer orders: small-scale (4 and 5 orders), medium-scale (6 and 8 orders), and large-scale (10, 12, and 14 orders). This section presents results from 20 small-scale, 15 medium-scale and 15 large-scale test instances. The LINGO model, along with the results and solution details generated by the ACS-HMLO framework, is available for download from⁶². The code of ACS-HMLO in C# can be obtained from⁶³.

Optimum performance (small-scale)

The 20 small-scale instances' numerical results are displayed in (Table 4). Problem instances column indicates the size of the evaluated problems, as “*k*” donates the number of machines, “*h*” donates the number of vehicles, “*n*” donates the number of orders, and *i* is the index of each problem. Table headers reading "Max," "Min," "Average," and "St. dev." indicate the highest and lowest solutions, average distribution cost, and standard deviation, as determined using the ACS-HMLO framework, respectively. The optimal distribution cost as determined by LINGO is denoted by *F*^{*}, and the least distribution cost as determined by the ACS-HMLO framework is denoted by *F*. *RD*₁ is calculated. The solving times for ACS-HMLO and LINGO, expressed in minutes, are displayed at the end of the (Table 4). Solving time for LINGO increases markedly as instance complexity grows, reaching up to 6.68 min for relatively larger instances (e.g., *k* = 2, *h* = 2, *n* = 5, *i* = 4). This highlights the computational overhead associated with exact methods in solving combinatorial optimization problems, which can be a limitation for real-time or large-scale applications. In contrast, the ACS-HMLO framework consistently exhibits significantly lower solving times, even as problem size increases. For the same large-scale instance, ACS-HMLO solves the problem in just 0.34 min, showcasing its efficiency in handling complex problems. This pronounced difference

Problem instances				Objective by ACS-HMLO						Solving time (minutes)	
<i>k</i>	<i>h</i>	<i>n</i>	<i>i</i>	Max	Min	Average	St. dev	LINGO	<i>RD</i> ₁ (%)	LINGO	ACS-HMLO
2	2	4	1	2452	2452	2452	0	2452	0.00	0.29	0.03
			2	1169	1169	1169	0	1169	0.00	0.24	0.03
			3	2078	2078	2078	0	2078	0.00	0.29	0.03
			4	689	689	689	0	689	0.00	0.07	0.03
			5	1002	1002	1002	0	1002	0.00	0.24	0.03
			6	1156	1156	1156	0	1156	0.00	0.13	0.03
			7	974	974	974	0	974	0.00	0.17	0.03
			8	1325	1325	1325	0	1325	0.00	0.14	0.03
			9	1645	1645	1645	0	1645	0.00	0.31	0.03
			10	784	784	784	0	784	0.00	0.31	0.03
		5	1	706	706	706	0	706	0.00	2.00	0.35
			2	1079	1041	1045.7	11.42	1041	0.00	1.66	0.34
			3	571	539	542.2	9.6	539	0.00	0.68	0.34
			4	1659	1659	1659	0	1659	0.00	6.68	0.34
			5	447	447	447	0	447	0.00	0.04	0.36
			6	1521	1521	1521	0	1521	0.00	6.35	0.33
			7	508	466	482.8	11	466	0.00	0.19	0.36
			8	618	618	618	0	618	0.00	0.15	0.36
			9	1055	1055	1055	0	1055	0.00	2.80	0.34
			10	417	388	400.5	12.58	388	0.00	0.19	0.44

Table 4. Numerical results for the 20 small-scale instances.

underscores the computational advantage offered by heuristic approaches such as ACS-HMLO, which achieve optimal solutions in all tested small-scale instances with much faster solving times.

While LINGO guarantees optimality through exhaustive search, its high computational cost can make it impractical for time-sensitive or resource-constrained scenarios. The ACS-HMLO framework, on the other hand, combines reliability with efficiency, offering a practical solution for small-scale IPS-VRTW problems where rapid decision-making is crucial. Optimal solutions for the 20 small-scale test instances (including up to 5 items, 2 machines, and 2 vehicles) can be found using the ACS-HMLO framework, according to the numerical results. The ACS-HMLO framework consistently finds optimal solutions for all small-scale cases. Furthermore, we have zero relative deviation (RD_1) across the 20 instances. Thus, it can be inferred that the ACS-HMLO framework is highly effective in solving small-scale IPS-VRTW problems.

Performance evaluation on medium-scale

To compare the solution results for medium and large-scale instances, we utilized the results from the LINGO software, as no prior study has addressed the problem with identical dimensions. The solution time for LINGO was constrained as follows: for the instances with eight orders, we set the solution time to 6 h; For instances with 8 orders, the solution time was established at 6 h; For 10 orders, it was set at 12 h; for 12 orders, a full day was allocated; and for 14 orders, the time was set as 3 days.

This subsection presents the performance evaluation of ACS-HMLO on medium-scale instances. A total of fifteen medium-scale instances were considered, consisting of 6 or 8 orders, 2 vehicles, and 2 machines. The goal is to validate the capability of ACS-HMLO to maintain computational efficiency and solution quality as the problem size increases.

The results, summarized in (Table 5), show the minimum distribution cost achieved by ACS-HMLO for each instance compared to the best solution found by LINGO. The RD_2 is calculated to quantify the improvement achieved by ACS-HMLO. In all cases, ACS-HMLO produced equal or superior objective values while significantly reducing computation time.

The alignment between the lowest values and the solutions provided by LINGO, combined with negative RD_2 values demonstrate that ACS-HMLO not only maintained optimality in all 6-order instances but also delivered consistent improvements in the 8-order cases, with relative deviations reaching up to -12.55%. Moreover, the computation time for ACS-HMLO remained under two minutes across all medium-scale problems, highlighting its potential for time-sensitive applications. These findings confirm that the proposed ACS-HMLO framework scales effectively and retains solution quality and speed as problem complexity increases moderately.

Performance evaluation on large-scale

Additionally, we evaluate the ACS-HMLO framework’s performance on 15 large-scale instances. The LINGO cannot find an optimal solution within a reasonable time. The comparison in this scale is based on the running CPU time and the quality of the best solution found. Results of large-scale instances and solving time in hours are shown in (Table 6). RD_3 is used to assess the gap between LINGO and ACS-HMLO for large-scale problems. The alignment between the lowest values and the solutions provided by LINGO, combined with negative RD_3 values, underscores the effectiveness of ACS-HMLO in achieving better or comparable solutions across various problem instances. The negative RD_3 values across the instances demonstrate that the ACS-HMLO framework not only provides solutions that are close to optimal, but in many cases, it outperforms LINGO, achieving lower objective values. This is particularly evident in instances such as ($k=2, h=2, n=10, i=1$), where ACS-HMLO yields a solution with a -13.29% RD_3 , indicating that the framework has successfully found a better solution

problem instances				Objective by ACS-HMLO				Solving time (minutes)			
k	h	n	i	Max	Min	Average	St. dev	LINGO	$RD_2(\%)$	LINGO	ACS-HMLO
2	2	6	1	2904	2856	2889.6	22	2856	0.00	64.56	1.49
			2	2448	2437	2440.8	4.69	2437	0.00	41.17	1.77
			3	4662	4520	4629.1	43.98	4520	0.00	42.57	1.47
			4	2200	2191	2199.1	2.7	2191	0.00	45.70	1.63
			5	4909	4909	4909	0	4909	0.00	132.07	1.57
			6	3758	3693	3706	26	3693	0.00	82.12	1.60
			7	2403	2403	2403	0	2403	0.00	42.61	1.55
			8	2993	2793	2859.8	56.46	2793	0.00	42.61	1.62
			9	3014	3014	3014	0	3014	0.00	37.92	1.63
			10	1199	1199	1199	0	1199	0.00	38.91	1.62
		8	1	1473	1377	1447.4	34.47	1379	-0.15	360.00	0.022
			2	1755	1601	1659.8	54.83	1751	-8.57	360.00	0.02
			3	1697	1610	1674.9	31.16	1615	-0.31	360.00	0.018
			4	1304	1298	1302.8	2.4	1367	-5.05	360.00	0.019
			5	961	892	922	29.09	1020	-12.55	360.00	0.018

Table 5. Numerical results for the 15 medium-scale instances.

Problem instances			Objective by ACS-HMLO						Solving time (hours)		
k	h	n	i	Max	Min	Average	St. dev	LINGO	$RD_3(\%)$	LINGO	ACS-HMLO
2	2	10	1	2215	1958	2027.6	90.43	2258	-13.29	12.00	0.004
			2	4003	3561	3779.9	114.31	5101	-30.19	12.00	0.005
			3	4472	3710	4228.2	235	4902	-24.32	12.00	0.005
			4	2719	2197	2484.4	140.35	3901	-43.68	12.00	0.004
			5	1963	1535	1802.2	141.67	2442	-37.14	12.00	0.003
		12	1	4876	4321	4611.7	141.91	6418	-32.67	22.87	0.005
			2	5753	5247	5516.4	210.3	6620	-20.74	23.00	0.007
			3	3911	3341	3586	152.85	4485	-25.51	23.03	0.007
			4	3021	2562	2834.1	129.31	3063	-16.36	24.00	0.006
			5	3254	2753	3039.2	161.34	4415	-37.64	24.00	0.006
		14	1	8923	8065	8530.3	245.62	8787	-8.22	92.71	0.01
			2	5564	4961	5323.7	168.45	8102	-38.77	93.42	0.012
			3	4788	4006	4366.6	222.88	8038	-50.16	93.36	0.012
			4	6260	5446	5810.4	267.55	7780	-30.00	93.32	0.009
			5	8460	7245	7838.8	340.06	8727	-16.98	72.84	0.008

Table 6. Numerical results for the 15 large-scale instances.

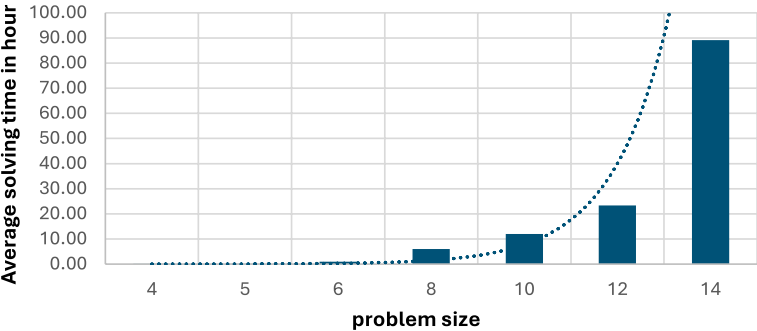


Fig. 5. Average solving time of LINGO for problems of different sizes(2–2- x).

than LINGO. Similarly, in larger problem instances like ($k=2, h=2, n=14, i=2$), the RD_3 value of -38.77% highlights a significant advantage of ACS-HMLO, where it achieves a substantially lower solution value compared to LINGO. The consistency of ACS-HMLO's performance is further reflected in the low standard deviations, which implies that the framework is stable and repeatable. While there is some variability in performance for larger problem instances, as evidenced by the higher standard deviations in those cases, the framework still maintains consistent solution quality and outperforms LINGO in many cases. Moreover, the solving times for ACS-HMLO continue to demonstrate its computational efficiency. For all problem instances, ACS-HMLO consistently achieves significantly faster solving times compared to LINGO, even for the larger instances. This stark difference highlights the computational advantage of ACS-HMLO, which provides a practical and fast solution, even as the problem scale increases. The time required for ACS-HMLO to solve large problem instances demonstrates its suitability for real-time or large-scale applications, where time constraints are crucial.

To determine the extent to which the size of the problem affects the solving time, Fig. 5 is developed. It shows the average solving time versus the problem size for a scenario with two machines, two vehicles, and a varying number of orders (x). The increase in the number of orders correlates with an exponential rise in the average solving time. For small problem sizes (up to about eight orders), the solving time is low. However, as the number of orders exceeds 10, the solving time rises sharply, reaching a remarkably high value at 14 orders. This suggests that the difficulty of solving the problem increases substantially with the addition of more orders. Table 6 and Fig. 5 demonstrate that the ACS-HMLO framework outperforms LINGO in terms of solution quality while requiring considerably less time for large-scale instances.

Figure 6 shows the average percentage of improvement using ACS-HMLO for different problem sizes, with two machines, two vehicles, and a varying number of orders (x). For small-scale instances (4 to 5 orders), the improvement is zero. This means that ACS-HMLO and LINGO will find the same solutions with the same objectives. For eight orders, the improvement is about 5%. For larger problem sizes (10, 12, and 14 orders), the improvement jumps to around 30%. This indicates that ACS-HMLO outperforms LINGO as the problem size increases.

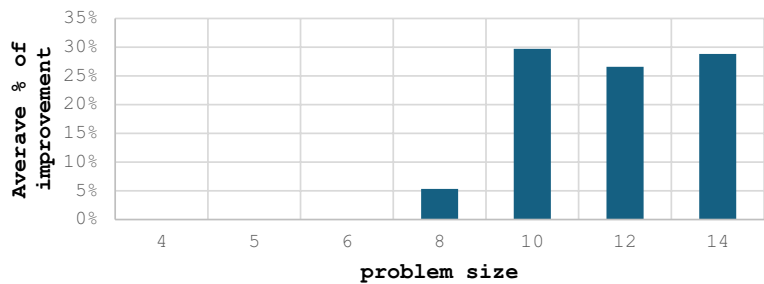


Fig. 6. Average % of improvement using ACS-HMLO for problems of different sizes (2–2-x).

Instance scale	Test used	Sample size	Test statistic	ρ_{value}	Significance
Small-scale (n = 20)	Not applicable (identical results)	20	–	–	No difference
Medium-scale (n = 15)	Not applicable (minimal deviation)	15	–	–	No meaningful difference
Large-scale (n = 15)	Paired-sample t-test	15	$t = -5.02$	< 0.001	Significant improvement

Table 7. Summary of statistical comparison of ACS-HMLO vs. LINGO.

Statistical validation of results

To confirm the robustness and significance of the performance improvements achieved by the proposed ACS-HMLO framework compared to the LINGO solver, formal statistical tests were conducted. The objective values (i.e., total distribution costs) obtained from ACS-HMLO and LINGO were paired and analyzed. Based on the characteristics of the data in each group, the following procedures were applied:

Small- and medium-scale instances For the small-scale instances, ACS-HMLO produced identical objective values to those obtained by LINGO. In the medium-scale cases, ACS-HMLO either matched or slightly outperformed LINGO. Due to the minimal or zero deviations across these instances, no statistical tests were performed for these categories, as the differences were not sufficient to provide meaningful statistical inferences.

Large-scale instances For the large-scale group, ACS-HMLO showed consistent improvement over LINGO. To determine whether these improvements were statistically significant, a normality check on the paired differences was first conducted using the Ryan-Joiner test (as implemented in Minitab). The test results yielded a correlation statistic of $R_j = 0.955$ and a $p - value$ of 0.074, indicating no significant departure from normality at $\alpha = 0.05$. Accordingly, a paired-sample t-test was applied to determine the statistical significance of the performance improvements. Summary of statistical comparison shown in Table 7 and the full test details are reported in Supplementary Note 3 online.

The test statistic $t = -5.02$ and its associated $\rho_{value} < 0.001$ indicate that the mean cost difference is statistically significant. Thus, the null hypothesis is rejected, confirming that ACS-HMLO significantly outperforms LINGO in large-scale instances. These findings indicate that the proposed ACS-HMLO framework is highly reliable, thus validating the effectiveness and scalability of the proposed approach. Full statistical outputs, including the Ryan-Joiner test and paired t-test details, are provided in Supplementary Note 3 online.

Sensitivity analysis

The impact of increasing machine numbers and the effectiveness of flexible departure time procedures

This section discusses the impact of increasing the number of machines on the total distribution cost in two scenarios: the first is without using the optimal departure time procedure, and the second is with using the optimal departure time procedure. Table 8 displays the numerical results for solving a problem involving 30 orders, two vehicles, and an increasing number of machines from 2 to 9.

Ignoring the use of the optimal departure time procedure generally results in notable enhancements in objective function values with an increase in the number of machines. The data indicates that objective values consistently decline with an increase in the number of machines up to a specific level. The increasing number of machines reduces order completion times and leads to earlier vehicle departure times, thereby decreasing tardiness. Increasing the number of machines will result in earlier times, subsequently raising the total distribution cost relative to the prior scenario. The improvement percentage will reach an optimal level. The analysis indicates that the optimal number of machines, in the absence of the optimal departure time procedure, is 4. The region exhibiting the lowest total distribution cost and the highest percentage of improvement is illustrated in (Fig. 7). The implementation of the optimal departure time procedure in various scenarios leads to notable enhancements in objective function values with an increase in the number of machines. The data analysis indicates a trend of decreasing minimum values across configurations as the number of machines increases from 2 to 9. The increasing number of machines gradually eliminates tardiness times while concurrently introducing earliness times. If the early time intervals increase significantly, the vehicle will encounter delays when utilizing

problem instances			Without using optimal dep. Time procedure					With using optimal dep. Time procedure	
<i>n</i>	<i>h</i>	<i>k</i>	Max	Min	Average	St. dev	% improvement	Distribution cost	% improvement
30	2	2	11407	7845	9212.9	979.2	0%	7845	0%
		3	6700	4396	5945.5	620.27	44%	4382	44%
		4	5979	3777	5306.9	741.24	52%	3699	53%
		5	5631	4844	5212.4	262.44	38%	4808	39%
		6	5369	4575	5046.8	289.55	42%	4575	42%
		7	5765	4515	5286.9	351.67	42%	4398	44%
		8	5638	4564	5243.8	301.75	42%	4104	48%

Table 8. Numerical results for solving the problem of 30 orders, two vehicles, and increasing number of machines.

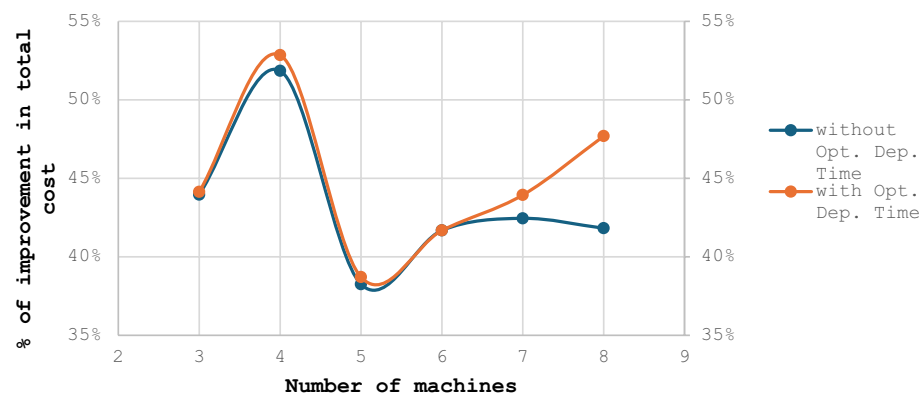


Fig. 7. The impact of increasing machines number on improvement percentage in a problem of 30 orders and 2 vehicles.

problem instances			Departure of vehicle 1		Departure of vehicle 2		avg. act dep	avg. opt dep
<i>n</i>	<i>h</i>	<i>k</i>	Actual	Optimal	Actual	Optimal		
30	2	2	166	166	367	367	267	267
		3	146	160	246	246	196	203
		4	102	102	188	199	145	151
		5	86	86	166	199	126	143
		6	59	59	149	149	104	104
		7	58	58	115	151	87	105
		8	47	108	115	115	81	112
		9	43	66	101	101	72	84

Table 9. Actual and optimal departure times of the two vehicles.

the optimal departure time method. Table 9 shows the optimization of the vehicle departure time with the aim of reducing the total distribution cost.

Figure 8 shows the average departure times of the two vehicles in both scenarios. The analysis indicates that increasing the number of machines, in the absence of an optimal departure time procedure, results in a significant reduction in the average actual departure time, consequently leading to elevated earliness costs. The optimal departure procedure restricts the average optimal departure time to mitigate the impact of earliness associated with an increase in the number of machines. The results presented in Table 8 and Fig. 7 show the effectiveness of using the optimal departure time procedure. The procedure resulted in lower costs and a higher percentage of improvement.

The impact of time window on total distribution cost

This analysis aims to assess the effects of altering time windows on the performance of the proposed ACS-HMLO. Two factors were assessed: the expansion of the time window width and the temporal shift of the time

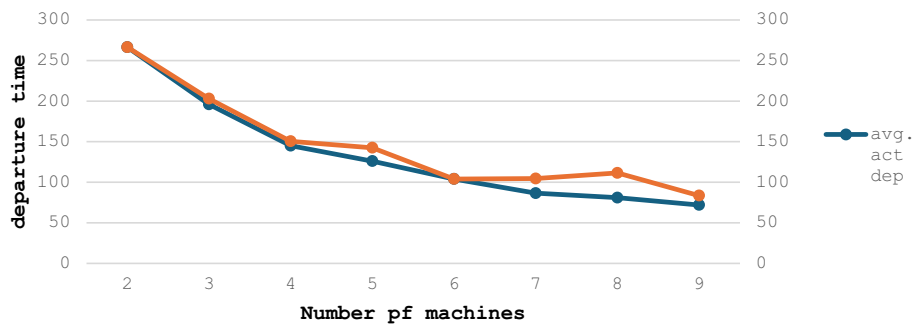


Fig. 8. Average departure time of the two vehicles in the two scenarios.

problem instances				Increasing width					Shifting time window				
n	h	k	% increase	Max	Min	Average	St. dev	% improvement	Max	Min	Average	St. dev	% improvement
30	2	2	0%	12121	8187	10254.3	1141.91	0%	12121	8187	10254.3	1141.91	0%
			10%	11280	7822	9507.5	964.52	4%	9017	6230	7424.3	844.61	24%
			20%	10058	7237	8581.4	864.41	12%	6673	3167	5332	947.77	61%
			30%	10040	5109	7842.3	1174.82	38%	6180	2965	4666.9	972.7	64%
			40%	7795	4846	6596.7	929.12	41%	5155	3514	4186.1	516	57%
			50%	7024	4750	5767.1	645.22	42%	3894	2557	3344	364.21	69%
			60%	8427	4483	6326.3	1101.4	45%	3540	2088	3060.9	466.35	75%
			70%	7782	3499	5830.1	1131.75	57%	3777	2024	2912.8	481.68	75%
			80%	5709	2624	4609	850.91	68%	3292	1641	2247.4	458.8	80%
			90%	5736	3577	4380.9	644.35	56%	2483	1723	2155.1	235.62	79%
			100%	4798	3002	3856	518.8	63%	2576	1522	2077.4	323.07	81%

Table 10. Numerical results for solving the problem of 30 orders, two vehicles, two machines, and increasing % in the time window.

window, each adjusted in 10% increments up to 100%. The numerical results of the two factors, as well as the percentage of improvement, are summarized in (Table 10). In the first test, increasing the width resulted in an average performance improvement of up to 68%. The second test demonstrated that altering the time window resulted in a significant average performance improvement, achieving up to 81%, which suggests enhanced stability. As shown in Fig. 9, adjusting the time window can yield a significant enhancement with only a 20% shift, while expanding the width requires an estimated 70% increase. This visual representation supports the conclusion that adjusting the time window is more effective and practical for improving the performance of integrated production scheduling and VRP problems.

Comparative analysis with related studies

Recent research on IPS-VRP demonstrates a shared recognition of the interdependence between manufacturing and distribution stages. Across all studies reviewed, the production environment involves parallel machines, either identical or heterogeneous, and the delivery stage incorporates vehicle routing with time window constraints. For large-scale problem instances, there is a common reliance on metaheuristic or hybrid approaches, as exact solvers such as CPLEX or LINGO become computationally infeasible. Moreover, most methods employ hybridization strategies to balance global exploration with local exploitation, reflecting a convergence in the search for efficient and high-quality solutions in complex, real-world contexts.

Despite these shared features, the studies differ substantially in their specific problem formulations, optimization objectives, and algorithmic designs. Our proposed ACS-HMLO framework is distinct in combining a two-phase decomposition approach with ACS metaheuristics, explicitly incorporating batch delivery, flexible departure times, and earliness/tardiness penalties to minimize total distribution costs. Sugianto and Kim²³ adopt Iterated Variable Neighborhood Search for additive manufacturing, emphasizing completion time minimization with sequence-dependent setups and independent production/delivery batches. Arda, et al.²² integrate Large Neighborhood Search with Linear Programming for home chemotherapy delivery, optimizing personnel working durations under strict stability and no-inventory constraints. Yağmur and Kesen²⁴ apply Iterated Local Search and Memetic Algorithms to an energy-aware scheduling problem, considering discrete machine speeds and load-dependent fuel consumption. Wu, et al.²⁵ focus on minimizing delivery delays in problems with sequence-dependent setups, using hybrid Variable Neighborhood Search combined with Cuckoo Search or Particle Swarm Optimization. These methodological and contextual differences shape the strengths

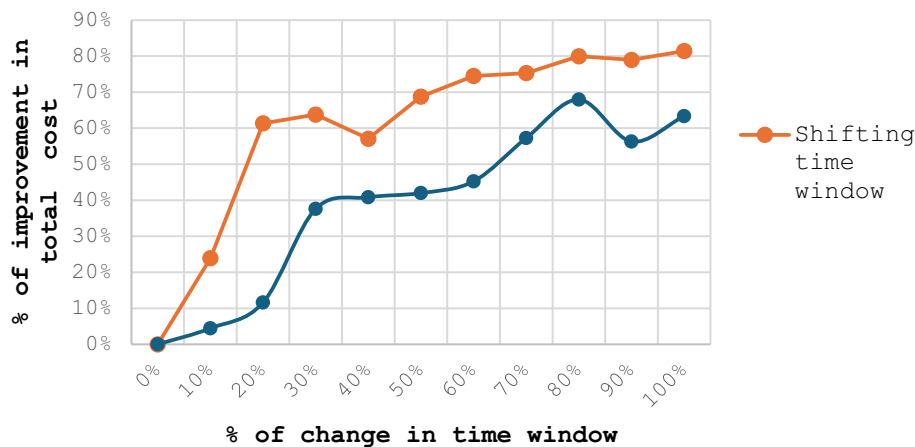


Fig. 9. The impact of change in the time window using two methods: increasing the width and shifting the time window.

Study	Problem context	Machines	Delivery features	Objective	Algorithm and distinctive features
Arda, et al. ²²	Home chemotherapy delivery	Heterogeneous machines (nurses)	Fixed departure, no-inventory constraint	Minimize personnel working duration	Large neighborhood search + Linear programming; stability constraint; medical delivery context
Sugianto and Kim ²³	Additive manufacturing scheduling & delivery	Parallel identical machines	Independent production/ delivery batches	Minimize completion time	Iterated Variable Neighborhood Search; Sequence-dependent setups; independent batching
Yağmur and Kesen ²⁴	Energy-aware scheduling & VRP	Parallel heterogeneous machines	Load-dependent fuel consumption	Minimize makespan & energy cost	Iterated local search + Memetic algorithm; Discrete machine speeds; energy optimization
Wu, et al. ²⁵	Production scheduling & VRP	Parallel identical machines	Sequence-dependent setups	Minimize delivery delays	Hybrid VNS + Cuckoo Search / PSO; Alternative metaheuristics for different problem scales
This study	Integrated production scheduling & VRP	Parallel identical machines	Batch delivery, flexible departure, earliness/tardiness penalties	Minimize total distribution cost	Two-phase ACS-HMLO; Two-phase decomposition; batch delivery with flexible departure times

Table 11. Comparative summary of IPS-VRP studies: problem settings, objectives, and algorithmic features.

and trade-offs observed, with the ACS-HMLO framework demonstrating particular advantages in balancing solution quality and computational efficiency for larger problem sizes.

Across the five studies, results consistently show that exact methods (e.g., MILP, MIP solved by CPLEX, Gurobi, LINGO) are effective for small-scale instances, achieving optimal solutions but with solving times that can rise rapidly with problem complexity. In contrast, metaheuristic approaches (e.g., LNS, ILS, MA, IVNS, VNS-CS, VNS-PSO, ACS-HMLO) solve these small problems within seconds, often faster than exact methods, while still reaching optimality or near-optimality. For medium- and large-scale instances, exact solvers become computationally impractical, often failing to find feasible solutions within hours or days. Metaheuristics, however, maintain scalability, producing high-quality or even superior solutions in significantly shorter times. Among them, VNS-based and ACS-HMLO frameworks consistently outperform others in both speed and solution quality, demonstrating their suitability for real-world integrated scheduling and routing problems. * MERGEFORMAT Table 11 summarizes the core characteristics of each study, detailing the problem context, machine configurations, delivery features, optimization objectives, and the distinctive elements of their algorithmic approaches. This structured comparison facilitates a clear understanding of how different problem settings and methodological choices influence solution strategies and performance outcomes.

The reviewed studies reveal that integrated production and distribution problems exhibit diverse sensitivity patterns to changes in key parameters, with both problem-specific characteristics and broader scaling effects influencing solution quality and computational efficiency. Several works highlight positive sensitivity to increased production resources, where adding machines often improves performance, though diminishing returns or earliness penalties may arise beyond optimal levels. Customer-related parameters, such as the number of customers, frequently demonstrate strong positive correlations with problem difficulty, substantially increasing total cost and computational time. Time-related parameters, including time window width, temporal positioning, and stability constraints, emerge as critical drivers of performance, with restrictive or poorly aligned time windows consistently amplifying costs, while wider or strategically shifted windows yield substantial improvements. Sensitivity is also observed in algorithmic design choices, such as the timing of exact optimization triggers, encoding methods, and integration strategies, all of which can significantly alter feasibility, solution quality, and computation time. * MERGEFORMAT Table 12 summarizes these findings, presenting for each study the specific parameters analyzed, their observed effects on performance, and the corresponding sensitivity patterns or implications. This comparative view highlights both common trends, such as the benefits

Study	Parameter analyzed	Observed effect	Sensitivity pattern/implication
Arda, et al. ²²	Integration vs. sequential planning	Integration increases administration duration by 15%	Necessary despite higher cost to meet stability and inventory constraints
	Drug stability times (STi)	Shorter stability times increase administration time by 4%	High sensitivity: stronger impact with short STi and longer horizons
	Administration time windows	Narrower windows sharply increase administration time, especially with short STi	High sensitivity: restrictive windows amplify costs
Sugianto and Kim ²³	Number of machines	Improved solution quality (lower RPD) with more machines when using the CUMax rule	Positive sensitivity: additional resources improve performance
	Machine area capacity (acm)	Excessively large capacities increase completion times and reduce efficiency	Optimal capacity range; benefits diminish beyond it
	Number of customers	RMin rule performs better with fewer customers and large machine capacities	Strong sensitivity: fewer customers simplify routing
	Overall robustness	IVNS-CUMax maintains stable performance across varied instance sizes	Robust except in extreme parameter ranges
Yağmur and Kesen ²⁴	Number of customers	Large increases in total cost and CPU time beyond 7 customers	Strong positive correlation: more customers increase complexity
	Number of machines	Higher machine count reduces total cost	Inverse sensitivity: more resources improve efficiency
	Number of vehicles	More vehicles increase total cost due to higher distribution energy costs	Positive sensitivity: reduced tardiness but higher travel costs
	Time window width (TW)	Wider windows greatly reduce cost and computational time	Strong inverse sensitivity to restrictiveness
Wu, et al. ²⁵	Problem size (jobs, machines, vehicles)	Performance gap between VNS-CS and VNS-PSO narrows as problem size increases	Scaling sensitivity: larger problems reduce performance differences
	Encoding method	Real encoding generally best; performance depends on matching encoding to algorithm	Sensitivity to representation choice; no universal best
This study	Number of machines	Costs decrease initially, then may rise due to earliness beyond optimal machine count	Diminishing returns; optimal number exists
	Flexible departure time	Up to 48% cost improvement with optimized departure timing	High sensitivity: precise timing improves performance
	Time WINDOW WIDTH EXPANSION	Up to 68% improvement from wider time windows	Strong positive correlation: flexibility reduces penalties
	Time window temporal shift	Up to 81% improvement from shifting windows strategically	Positioning more effective than width expansion

Table 12. Sensitivity of integrated production and distribution problems to changes in problem parameters.

of increased production resources and the adverse impact of restrictive time windows, and problem-specific sensitivities that necessitate tailored algorithmic approaches.

Managerial insights and policy implications

The results of the sensitivity analyses conducted in this study provide several actionable insights for managers aiming to optimize integrated production scheduling and vehicle routing with time windows. Key findings highlight the significant impact of machine availability, flexible departure times and the impact of time window adjustments. on overall cost efficiency and service performance.

Optimizing machine utilization

Increasing the number of machines reduces production completion times, allowing earlier vehicle departures and improving overall delivery performance. However, beyond a certain threshold, additional machines contribute minimal further cost reductions and may lead to increased earliness penalties. Managers should evaluate the trade-off between machine investment costs and marginal cost savings to determine the optimal machine configuration for their operations.

Flexible departure times for cost efficiency

Implementing flexible departure time policies significantly lowers total distribution costs by balancing early and tardy deliveries. The study demonstrates that optimizing departure times reduces penalties and enhances service reliability. Managers can implement our procedure to optimize their vehicle departure schedules based on real-time production completion data rather than using a fixed schedule. This allows for better synchronization between production and delivery, minimizing the cost of time window violations.

Impact of time window adjustments

Expanding or shifting delivery time windows affects total distribution costs and service levels. Wider time windows provide more scheduling flexibility, reducing penalties and transportation expenses. Conversely, strategically shifting time windows can yield faster improvements in cost performance with fewer changes required. Managers can use ACS-HMLO to evaluate the optimal width and placement of time windows for critical customer segments to maximize efficiency.

Balancing computational resources and solution quality

The ACS-HMLO framework offers a practical trade-off between computational time and solution quality. For large-scale problems, heuristic-based methods provide near-optimal solutions within feasible time limits. Decision-makers in time-sensitive industries can prioritize ACS-HMLO for quick, reliable scheduling decisions.

Policy recommendations for different industries

For industries handling perishable goods, optimizing departure times using ACS-HMLO can reduce spoilage risks and ensure compliance with cold chain logistics requirements. In e-commerce, dynamic time window adjustments can enhance delivery accuracy while minimizing logistics costs. In manufacturing, adjusting machine allocations based on distribution parameters can enhance resource utilization and minimize operational costs. In healthcare and Pharmaceutical Supply Chains, efficient production–distribution integration ensures timely deliveries of critical medical supplies and medications. ACS-HMLO can minimize delivery deviations, reducing risks in emergency logistics for hospitals and clinics.

Conclusion and future work

The study effectively demonstrates the potential of an HMLO to address the integrated production and distribution scheduling problem at operational level. This framework integrates metaheuristics and heuristics to optimize the makespan and delivery times in a complex scheduling environment. The results from numerical experiments validate the framework's efficacy, showing significant improvements in both the production and distribution phases.

An MILP model is developed to solve the specified problem. The model aims to minimize total distribution costs, encompassing travel expenses and penalties for breaching designated time windows. The HMLO framework is designed to effectively manage the complexities associated with scheduling parallel machines and VRP-TW in real-world scenarios. The framework integrates multiple optimization techniques, utilizing the strengths of each method to provide a robust and flexible solution to the scheduling problem. The integration of the ACS with HMLO has demonstrated notable effectiveness, achieving a balance between solution quality and computational efficiency. The framework significantly reduces the makespan, enhancing overall production efficiency. Delivery times are optimized, ensuring timely deliveries and minimizing penalties associated with violating the delivery time window. The computational experiments highlight the framework's scalability, demonstrating its applicability to small-scale, medium-scale and large-scale scheduling problems. The ACS-HMLO framework outperforms traditional methods, such as those implemented in LINGO software, especially in medium and large-scale instances where LINGO struggles to find optimal solutions within a reasonable time. The hybrid approach not only provides high-quality solutions but also does so in a fraction of the time required by conventional methods. Sensitivity analysis is also constructed to measure the impact of increasing machine numbers and changing the time window by increasing the width or shifting the time window. The effectiveness of using optimal departure time procedures is measured while measuring the impact of increasing machines' numbers, as the two factors are closely related.

This study successfully addressed the three research questions posed in the introduction. First, the proposed HMLO framework effectively integrates production scheduling and vehicle routing, demonstrating significant cost reductions and improved operational efficiency. Second, the ACS-HMLO algorithm outperforms traditional exact methods such as LINGO in terms of computational time and solution quality, particularly for large-scale instances. Finally, sensitivity analysis confirmed that key parameters, such as machine numbers and flexible departure times, significantly influence cost efficiency and robustness. These findings highlight the practical applicability of ACS-HMLO in optimizing integrated supply chain operations.

The comparative analysis with recent IPS-VRP studies highlights that the proposed ACS-HMLO framework achieves competitive or superior performance compared to state-of-the-art metaheuristic methods, particularly in large-scale instances where exact solvers become infeasible. The sensitivity evaluation further underscores the robustness of the framework, demonstrating consistent performance gains across diverse operational settings, including flexible departure times and optimized time windows. These findings confirm the method's scalability, adaptability, and practical relevance in integrated production and distribution environments.

In conclusion, the proposed HMLO framework represents a significant advancement in the field of production and distribution scheduling. This work addresses significant challenges associated with integrating these two phases, offering a comprehensive solution that improves efficiency and reduces costs. The study's findings suggest that adopting such hybrid optimization techniques can lead to substantial operational improvements and cost savings in various industrial applications. Future work should focus on refining the HMLO framework to improve efficiency and adaptability, integrating advanced metaheuristics and machine learning algorithms for improved solution quality, and expanding its application to various industries to validate its generalizability. Additionally, developing capabilities for real-time scheduling and dynamic adjustment based on changing conditions will render the framework more applicable to real-world scenarios. The focus will also be on incorporating recent hybrid metaheuristics from 2022–2024, to validate our framework against methods explicitly designed for similar problem structures. Finally, future research should incorporate sustainability factors, including the reduction of energy consumption and carbon footprints, into the optimization framework.

Data availability

The LINGO model, along with the results and solution details generated by the ACS-HMLO framework for 20 small-scale, 15 medium-scale and 15 large-scale test instances, is available for download from ⁶². Code written in C# executes the ACS-HMLO algorithm and can be obtained from ⁶³.

Received: 22 April 2025; Accepted: 19 September 2025

Published online: 22 November 2025

References

- Moons, S., Ramaekers, K., Caris, A. & Arda, Y. Integration of order picking and vehicle routing in a B2C e-commerce context. *Flex. Serv. Manuf. J.* **30**, 813–843. <https://doi.org/10.1007/s10696-017-9287-5> (2018).
- Iyer, A. V., Seshadri, S. & Vasher, R. *Toyota Supply Chain Management: A Strategic Approach to the Principles of Toyota's Renowned System* (McGraw-Hill, 2009).
- Moons, S., Ramaekers, K., Caris, A. & Arda, Y. Integrating production scheduling and vehicle routing decisions at the operational decision level: A review and discussion. *Comput. Ind. Eng.* **104**, 224–245. <https://doi.org/10.1016/j.cie.2016.12.010> (2017).
- Yağmur, E. & Kesen, S. E. A memetic algorithm for joint production and distribution scheduling with due dates. *Comput. Ind. Eng.* **142**, 106342. <https://doi.org/10.1016/j.cie.2020.106342> (2020).
- Mohammadi, S., Al-e-Hashem, S. M. J. M. & Rekik, Y. An integrated production scheduling and delivery route planning with multi-purpose machines: A case study from a furniture manufacturing company. *Int. J. Prod. Econ.* **219**, 347–359. <https://doi.org/10.1016/j.iipe.2019.05.017> (2020).
- Bard, J. F. & Nananukul, N. The integrated production–inventory–distribution–routing problem. *J. Sched.* **12**, 257–280. <https://doi.org/10.1007/s10951-008-0081-9> (2009).
- Boudia, M., Dauzere-Peres, S., Prins, C. & O. Louly, M. Integrated optimization of production and distribution for several products in *International Conference on Services Systems and Services Management, ICSSSM*. 272–277 <https://doi.org/10.1109/icsssm.2006.320625> (IEEE).
- Safra, I., Jebali, A., Jemai, Z., Bouchriha, H. & Ghaffari, A. The beneficial effect of information sharing in the integrated production–distribution planning of textile and apparel supply chain. *RAIRO – Operat. Res.* **55**, 1171–1195. <https://doi.org/10.1051/ro/2021038> (2021).
- He, P., Li, K. & Kumar, P. N. R. An enhanced branch-and-price algorithm for the integrated production and transportation scheduling problem. *Int. J. Prod. Res.* **60**, 1874–1889. <https://doi.org/10.1080/00207543.2021.1876941> (2022).
- Guarnaschelli, A., Salomone, H. E. & Méndez, C. A. A stochastic approach for integrated production and distribution planning in dairy supply chains. *Comput. Chem. Eng.* **140**, 106966. <https://doi.org/10.1016/j.compchemeng.2020.106966> (2020).
- Bilgen, B. & Ozkarahan, I. Strategic tactical and operational production-distribution models: a review. *Int. J. Technol. Manag.* **28**, 151–171. <https://doi.org/10.1504/ijtm.2004.005059> (2004).
- Fahimnia, B., Farahani, R. Z., Marian, R. & Luong, L. A review and critique on integrated production–distribution planning models and techniques. *J. Manuf. Syst.* **32**, 1–19. <https://doi.org/10.1016/j.jmsy.2012.07.005> (2013).
- Vidal, C. J. & Goetschalckx, M. Strategic production-distribution models: A critical review with emphasis on global supply chain models. *Eur. J. Oper. Res.* **98**, 1–18. [https://doi.org/10.1016/s0377-2217\(97\)80080-x](https://doi.org/10.1016/s0377-2217(97)80080-x) (1997).
- Sarmiento, A. M. & Nagi, R. A review of integrated analysis of production-distribution systems. *IIE Trans.* **31**, 1061–1074. <https://doi.org/10.1080/07408179908969907> (1999).
- Díaz-Madroño, M., Peidro, D. & Mula, J. A review of tactical optimization models for integrated production and transport routing planning decisions. *Comput. Ind. Eng.* **88**, 518–535. <https://doi.org/10.1016/j.cie.2015.06.010> (2015).
- Chen, Z.-L. Integrated production and outbound distribution Scheduling: review and extensions. *Oper. Res.* **58**, 130–148. <https://doi.org/10.1287/opre.1080.0688> (2010).
- Wang, D. Y., Grunder, O. & Moudni, A. E. Integrated scheduling of production and distribution operations: a review. *Int. J. Ind. Syst. Eng.* <https://doi.org/10.1504/ijise.2015.065949> (2015).
- Berghman, L., Kergosien, Y. & Billaut, J. C. A review on integrated scheduling and outbound vehicle routing problems. *Eur. J. Oper. Res.* **311**, 23. <https://doi.org/10.1016/j.ejor.2022.12.036> (2023).
- Elsaeed, A., Ashour, M. & Elshaer, R. Integrating production scheduling and vehicle routing decisions at the operational decision level: A review and discussion extension. *J. Adv. Manuf. Syst.* **23**, 409–436. <https://doi.org/10.1142/s0219686724500173> (2023).
- Tavares-Neto, R. F. & Nagano, M. S. An iterated greedy approach to integrate production by multiple parallel machines and distribution by a single capacitated vehicle. *Swarm Evol. Comput.* **44**, 612–621. <https://doi.org/10.1016/j.swevo.2018.08.001> (2019).
- Kergosien, Y., Gendreau, M. & Billaut, J. C. A Benders decomposition-based heuristic for a production and outbound distribution scheduling problem with strict delivery constraints. *Eur. J. Oper. Res.* **262**, 287–298. <https://doi.org/10.1016/j.ejor.2017.03.028> (2017).
- Arda, Y., Cattaruzza, D., François, V. & Ogier, M. Home chemotherapy delivery: An integrated production scheduling and multi-trip vehicle routing problem. *Eur. J. Oper. Res.* **317**, 468–486. <https://doi.org/10.1016/j.ejor.2024.03.039> (2024).
- Sugianto, W. C. & Kim, B. S. Iterated variable neighborhood search for integrated scheduling of additive manufacturing and multi-trip vehicle routing problem. *Comput. Operat. Res.* <https://doi.org/10.1016/j.cor.2024.106659> (2024).
- Yağmur, E. & Kesen, S. E. Integrated production scheduling and vehicle routing problem with energy efficient strategies: Mathematical formulation and metaheuristic algorithms. *Expert Syst. Applic.* <https://doi.org/10.1016/j.eswa.2023.121586> (2024).
- Wu, G. H., Cheng, C. Y., Pourhejazy, P. & Fang, B. L. Variable neighborhood-based Cuckoo Search for production routing with time window and setup times. *Appl. Soft Comput.* <https://doi.org/10.1016/j.asoc.2022.109191> (2022).
- Schubert, D., Kuhn, H. & Holzapfel, A. Same-day deliveries in omnichannel retail: Integrated order picking and vehicle routing with vehicle-site dependencies. *Naval Res. Logistics (NRL)* **68**, 721–744. <https://doi.org/10.1002/nav.21954> (2021).
- Liu, D., Wang, W., Huang, L. & Proverbs, D. Coordinated production and delivery operations with parallel machines and multiple vehicles. *IEEE Access* **8**, 32947–32956. <https://doi.org/10.1109/ACCESS.2019.2958742> (2020).
- Liu, H., Guo, Z. & Zhang, Z. A hybrid multi-level optimisation framework for integrated production scheduling and vehicle routing with flexible departure time. *Int. J. Prod. Res.* **59**, 6615–6632. <https://doi.org/10.1080/00207543.2020.1821927> (2021).
- Wang, D. et al. Integrated production and multiple trips vehicle routing with time windows and uncertain travel times. *Comput. Oper. Res.* **103**, 1–12. <https://doi.org/10.1016/j.cor.2018.10.011> (2019).
- Kesen, S. E. & Bektaş, T. Integrated production scheduling and distribution planning with time windows. In *Lean and Green Supply Chain Management International Series in Operations Research & Management Science* (eds Paksoy, T. et al.) (Springer International Publishing, 2019).
- Gharaei, A. & Jolai, F. A branch and price approach to the two-agent integrated production and distribution scheduling. *Comput. Ind. Eng.* **136**, 504–515. <https://doi.org/10.1016/j.cie.2019.07.050> (2019).
- Dayarian, I. & Desaulniers, G. A branch-price-and-cut algorithm for a production-routing problem with short-life-span products. *Transp. Sci.* **53**, 829–849. <https://doi.org/10.1287/trsc.2018.0854> (2019).
- Fu, L. L., Aloulou, M. A. & Triki, C. Integrated production scheduling and vehicle routing problem with job splitting and delivery time windows. *Int. J. Prod. Res.* **55**, 5942–5957. <https://doi.org/10.1080/00207543.2017.1308572> (2017).
- Zhong, X. & Jiang, D. Parallel machine scheduling with batch delivery to two customers. *Math. Probl. Eng.* **2015**, 247356. <https://doi.org/10.1155/2015/247356> (2015).
- Lee, J., Kim, B.-I., Johnson, A. L. & Lee, K. The nuclear medicine production and delivery problem. *Eur. J. Oper. Res.* **236**, 461–472. <https://doi.org/10.1016/j.ejor.2013.12.024> (2014).
- Chang, Y.-C., Li, V. C. & Chiang, C.-J. An ant colony optimization heuristic for an integrated production and distribution scheduling problem. *Eng. Optim.* **46**, 503–520. <https://doi.org/10.1080/0305215X.2013.786062> (2014).
- Ullrich, C. A. Integrated machine scheduling and vehicle routing with time windows. *Eur. J. Oper. Res.* **227**, 152–165. <https://doi.org/10.1016/j.ejor.2012.11.049> (2013).
- Chang, Y.-C., Chang, K.-H. & Chang, T.-K. Applied column generation-based approach to solve supply chain scheduling problems. *Int. J. Prod. Res.* **51**, 4070–4086. <https://doi.org/10.1080/00207543.2013.774476> (2013).

39. Belo-Filho, M. A. F., Amorim, P. & Almada-Lobo, B. An adaptive large neighbourhood search for the operational integrated production and distribution problem of perishable products. *Int. J. Prod. Res.* **53**, 6040–6058. <https://doi.org/10.1080/00207543.2015.1010744> (2015).
40. Amorim, P., Belo-Filho, M. A. F., Toledo, F. M. B., Almeder, C. & Almada-Lobo, B. Lot sizing versus batching in the production and distribution planning of perishable goods. *Int. J. Prod. Econ.* **146**, 208–218. <https://doi.org/10.1016/j.ijpe.2013.07.001> (2013).
41. Farahani, P., Grunow, M. & Günther, H. O. Integrated production and distribution planning for perishable food products. *Flex. Serv. Manuf. J.* **24**, 28–51. <https://doi.org/10.1007/s10696-011-9125-0> (2012).
42. Liu, H., Guo, Z. & Zhang, Z. A hybrid multi-level optimisation framework for integrated production scheduling and vehicle routing with flexible departure time. *Int. J. Prod. Res.* **59**, 6615–6632. <https://doi.org/10.1080/00207543.2020.1821927> (2020).
43. Lin, C. W., Lin, Y. K. & Hsieh, H. T. Ant colony optimization for unrelated parallel machine scheduling. *Int. J. Adv. Manuf. Technol.* **67**, 35–45. <https://doi.org/10.1007/s00170-013-4766-7> (2013).
44. Ganji, M., Kazemipoor, H., Hadji Molana, S. M. & Sajadi, S. M. A green multi-objective integrated scheduling of production and distribution with heterogeneous fleet vehicle routing and time windows. *J. Clean. Prod.* **259**, 1–18. <https://doi.org/10.1016/j.jclepr.2020.120824> (2020).
45. Ganji, M., Rabet, R. & Sajadi, S. M. A new coordinating model for green supply chain and batch delivery scheduling with satisfaction customers. *Environ. Dev. Sustain.* **24**, 4566–4601. <https://doi.org/10.1007/s10668-021-01620-9> (2021).
46. Dorigo, M. & Gambardella, L. M. Ant colony system: a cooperative learning approach to the traveling salesman problem. *IEEE Trans. Evol. Comput.* **1**, 53–66. <https://doi.org/10.1109/4235.585892> (1997).
47. Dorigo, M., Caro, G. D. & Gambardella, L. M. Ant algorithms for discrete optimization. *Artif. Life* **5**, 137–172. <https://doi.org/10.1162/106454699568728> (1999).
48. Montemanni, R., Gambardella, L. M., Rizzoli, A. E. & Donati, A. V. Ant colony system for a dynamic vehicle routing problem. *J. Comb. Optim.* **10**, 327–343. <https://doi.org/10.1007/s10878-005-4922-6> (2005).
49. Barán, B. & Schaerer, M. A Multiobjective ant colony system for vehicle routing problem with time windows in *Applied Informatics*. <https://api.semanticscholar.org/CorpusID:641885>
50. Donati, A. V., Montemanni, R., Casagrande, N., Rizzoli, A. E. & Gambardella, L. M. Time dependent vehicle routing problem with a multi ant colony system. *Eur. J. Oper. Res.* **185**, 1174–1191. <https://doi.org/10.1016/j.ejor.2006.06.047> (2008).
51. Dorigo, M. & Gambardella, L. M. Ant colonies for the travelling salesman problem. *Biosystems* **43**, 73–81. [https://doi.org/10.1016/S0303-2647\(97\)01708-5](https://doi.org/10.1016/S0303-2647(97)01708-5) (1997).
52. Dorigo, M., Maniezzo, V. & Colnari, A. Ant System: An Autocatalytic Optimizing Process 91–106 (Department of Electronic, Politecnico of Milano, Milan, Italy) <https://api.semanticscholar.org/CorpusID:18900660>
53. Bullnheimer, B., Hartl, R. F. & Strauß, C. A new rank based version of the ant system. A computational study. *Adapt. Inform. Syst. Modell. Econ. Manag. Sci.* <https://doi.org/10.57938/2eb99227-a596-4eb4-bb7d-367b53dee89d> (1997).
54. Stützle, T. & Hoos, H. Improvements on the ant-system: Introducing the MAX-MIN Ant system. In: *Artificial Neural Nets and Genetic Algorithms*. (eds George D. Smith, Nigel C. Steele, & Rudolf F. Albrecht) 245–249 (Springer, Vienna) https://doi.org/10.1007/978-3-7091-6492-1_54
55. Tavares Neto, R. F., Godinho Filho, M. & da Silva, F. M. An ant colony optimization approach for the parallel machine scheduling problem with outsourcing allowed. *J. Intell. Manuf.* **26**, 527–538. <https://doi.org/10.1007/s10845-013-0811-5> (2013).
56. Chang, Y. C., Li, V. C. & Chiang, C. J. An ant colony optimization heuristic for an integrated production and distribution scheduling problem. *Eng. Optim.* **46**, 503–520. <https://doi.org/10.1080/0305215x.2013.786062> (2013).
57. Gambardella, L. M., Taillard, E. & Agazzi, G. MACS-VRPTW: A Multiple Ant Colony System for Vehicle Routing Problems with Time Windows (Istituto Dalle Molle Di Studi Sull Intelligenza Artificiale, 1999).
58. Gharaei, A. & Jolai, F. A Pareto approach for the multi-factory supply chain scheduling and distribution problem. *Oper. Res. Int. J.* **21**, 2333–2364. <https://doi.org/10.1007/s12351-019-00536-7> (2019).
59. Izadi, L., Ahmadizar, F. & Arkat, J. A hybrid genetic algorithm for integrated production and distribution scheduling problem with outsourcing allowed. *Int. J. Eng.* **33**, 2285–2298. <https://doi.org/10.5829/ije.2020.33.11b.19> (2020).
60. An, Y. et al. A hybrid multi-objective evolutionary algorithm for solving an adaptive flexible job-shop rescheduling problem with real-time order acceptance and condition-based preventive maintenance. *Expert Syst. Appl.* **212**, 118711. <https://doi.org/10.1016/j.eswa.2022.118711> (2023).
61. Huang, M., Du, B. & Guo, J. A hybrid collaborative framework for integrated production scheduling and vehicle routing problem with batch manufacturing and soft time windows. *Comput. Operat. Res.* <https://doi.org/10.1016/j.cor.2023.106346> (2023).
62. Elsaedi, A., Ashour, M. & Elshaer, R. *Numerical results for LINGO and ACS-HMLO*. (https://engzuedu-my.sharepoint.com/:f/g/personal/amelsaeed_eng_zu_edu_eg/EurzjpINvhVKt1zQKpmUayYB4MnCdTavNPDdSOUnF-NbDA) (2025).
63. ACS-HMLO framework. <https://doi.org/10.5281/zenodo.14755215> (Zenodo, 2025).

Acknowledgements

The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project.

Author contributions

Conceptualization, A.E. and R.E.; methodology, A.E., M.A. and R.E.; software, A.E. and R.E.; validation, A.E., M.A. and R.E. formal analysis, A.E., M.A. and R.E.; data curation, A.E. writing—original draft preparation, A.E., M.A. Y.M. and R.E.; writing—review and editing, A.E., M.A. Y.M. and R.E.; visualization, A.E., M.A. Y.M. and R.E.; supervision, M.A. Y.M. and R.E. project administration, Y.M. and R.E.; funding acquisition, R.E. All authors have read and agreed to the published version of the manuscript.

Funding

The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project under Grant Number RGP2/513/46.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-025-21231-y>.

Correspondence and requests for materials should be addressed to A.E. or R.E.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025